

Nonuniqueness for the periodic hard-sphere Boltzmann equation

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Abstract

We prove nonuniqueness for the periodic hard-sphere Boltzmann equation by constructing two distinct global renormalized solutions with the same nonnegative initial density on $\mathbb{T}^3 \times \mathbb{R}^3$. This density has bounded velocity support and finite mass, energy, and absolute entropy. Both solutions conserve local mass and total momentum and satisfy the global energy and entropy-dissipation inequalities. On a common initial interval, they are strongly continuous in L^1 , and their collision gains and losses are integrable.

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1 Introduction

We study the Cauchy problem for the hard-sphere Boltzmann equation on $\Omega = \mathbb{T}_x^3 \times \mathbb{R}_v^3$, where $\mathbb{T}^3 = \mathbb{R}^3/\mathbb{Z}^3$. A nonnegative density $F(t, x, v)$ evolves by free transport and binary elastic collisions:

$$(\partial_t + v \cdot \nabla_x)F = Q(F, F), \quad F(0, x, v) = F_0(x, v).$$

For incoming velocities v, v_* and a collision normal $\omega \in S^2$, put

$$a = (v - v_*) \cdot \omega, \quad v' = v - a\omega, \quad v'_* = v_* + a\omega.$$

Our normalization is

$$Q(F, F)(v) = \int_{\mathbb{R}^3 \times S^2} |a| (F(v')F(v'_*) - F(v)F(v_*)) dv_* d\omega,$$

with the common variables t, x suppressed and ordinary surface area measure on S^2 . The initial density is assumed to have finite mass, kinetic energy, and absolute entropy, meaning that $\int_{\Omega} F_0(1 + |v|^2 + |\log F_0|) < \infty$, with $0|\log 0| = 0$. The question is whether these physical bounds and the renormalized equation determine the subsequent evolution uniquely.

The global existence and weak stability theory of DiPerna and Lions provides solutions at this level of integrability [3]. Their entropy-production inequality explicitly includes periodic space [4]; [11, Theorem 4.1] gives a periodic formulation with a Maxwellian reference density. The need for renormalization is substantive: the stated bounds do not themselves make the quadratic collision term locally integrable. Existence and weak stability do not supply uniqueness. The uniqueness problem for general renormalized solutions is discussed, for example, in [13, Section 2].

We use the following precise admissibility requirements, written in full in Definition 2.1. With $M(v) = (2\pi)^{-3/2}e^{-|v|^2/2}$, the equation is tested through $M\beta(F/M)$ for every C^1 function β satisfying $\beta(0) = 0$ and $|\beta'(z)| \leq C_\beta/(1 + |z|)$; the separately renormalized gain and loss are integrable on every finite time interval. The solution is weakly continuous in $L^1(\Omega)$, against every bounded measurable test, and takes the prescribed initial value. Its mass, energy, and absolute entropy remain bounded on bounded time intervals. It conserves local mass and total momentum and satisfies both the global energy inequality and the relative-entropy inequality with the full Boltzmann entropy dissipation at every time. Here admissibility does not require local momentum or local energy conservation. Our main result concerns uniqueness within this class.

Theorem 1.1. *There is a nonnegative density F_0 on $\mathbb{T}^3 \times \mathbb{R}^3$, with bounded velocity support and*

$$\int_{\mathbb{T}^3 \times \mathbb{R}^3} F_0(1 + |v|^2 + |\log F_0|) dx dv < \infty,$$

for which the hard-sphere Boltzmann equation has two distinct global admissible solutions, in the sense of Definition 2.1, with initial datum F_0 . On some common initial interval both solutions are strongly continuous in L^1 , and their collision gains and losses are integrable in space-time-velocity.

The two solutions therefore obey the unrenormalized equation on the interval where their difference is established. Their common datum has bounded velocities but unbounded amplitudes concentrated at successively smaller spatial scales. The conservation and dissipation requirements remain part of the conclusion.

1.1 Uniqueness and related constructions

Classical uniqueness theorems impose additional control, such as Maxwellian upper bounds [10]. For periodic cutoff hard potentials, Ukai’s perturbative theory gives global existence and uniqueness near a Maxwellian in a velocity-weighted spatial Sobolev class, with mass, momentum, and energy matched to that Maxwellian [14, Theorem 1.2].

The distinction between spatially homogeneous and inhomogeneous evolution is also important. For the homogeneous hard-sphere equation, Mischler and Wennberg prove uniqueness in a conservative class from finite mass and energy alone [12, Theorem 1.1]. Wennberg’s homogeneous nonuniqueness examples for hard potentials permit increasing energy [15]. Theorem 1.1 instead concerns a spatially inhomogeneous datum, with the energy and entropy-dissipation inequalities retained.

Recent instability constructions give another relevant comparison. Chen and Holmer prove failure of uniform continuity in weighted Sobolev spaces for the constant collision kernel [2, Theorem 1.3]; Chen, Guo, Shen, and Zhang establish failure of uniform continuity for the hard-sphere solution map on $\mathbb{R}_x^3 \times \mathbb{R}_v^3$, in Gaussian-weighted $L_v^2 H_x^s$ spaces for $0 \leq s < 1$ [1, Theorem 1.1]. Those results compare solutions with arbitrarily close initial data. Here the two initial data agree exactly, on periodic space, and both resulting solutions satisfy the admissibility conditions above. The instability in [1] is driven by collision loss and dispersion. In the construction below, collision gain amplifies a vanishing hot seed and produces a nonzero weak limit near the initial spatial singularity.

For the related Krieger–Strain model, Gismondi, Golding, and Novack construct rough stationary solutions in a mollifier-confluent sense: the collision operators of mollified densities tend to zero in distributions for every nonnegative Schwartz mollifier of mass one [7, Definition 1.3 and Theorem 1.5]. They also conjecture nonuniqueness of renormalized inhomogeneous Landau and Boltzmann solutions for a dense class of initial data. Theorem 1.1 concerns one periodic hard-sphere datum in the specified admissibility class.

1.2 One datum and two approximation families

Write $x = (y, z) \in \mathbb{R}^2 \times \mathbb{R}$ in a lift of the torus, and $v = (\bar{v}, v_3) \in \mathbb{R}^2 \times \mathbb{R}$. The datum consists of narrow velocity distributions, called cold jets, near the axial directions $\pm e_3$. They occupy transverse annuli of geometrically decreasing radii $s_j \downarrow 0$. Within each annulus the two velocity signs occupy disjoint, finely alternating stripes. Their thermal widths are much smaller than the stripe gaps. An expanding axial velocity profile bounds the spatial density of each freely transported jet at scale s by $C/(t + s)$. Annular mass is $O(s^2)$, and annular absolute entropy is $O(s^2(1 + |\log s|))$, so all scales together define a single finite-entropy datum.

Assign a color to each cold jet. A collision within one color retains that color; collisions between different colors, or with particles already declared hot, produce hot particles. This is an exact decomposition: the sum of the colored equations is the Boltzmann equation. The narrow velocity distributions suppress self-collisions, and the spatial gaps suppress collisions between distinct cold jets at early times.

Retain the first N annuli and impose a collision cutoff that is removed as $N \rightarrow \infty$. Let h_N denote the hot component. The estimates use an integrable velocity weight $H(v)$, Gaussian at infinity and locally comparable to $|\bar{v}|^{-q}$ near the velocity axis, with $1 < q < 2$. A *cap* is an almost-everywhere upper bound on h_N/H . The two approximation families have the following properties:

1. With no hot particles initially, the hot density remains bounded by $t^2 H(v)$. We call this the dormant family.
2. For every sufficiently short observation horizon $T > 0$, a smooth hot seed of vanishing mass, energy, and entropy can be selected so that $h_N \leq (t + s_N)H$ on $[0, T]$, with equality in the weighted essential supremum at some time. Here s_N is the smallest retained annular radius.

At each fixed horizon, uniform integrability of both the densities and their two-particle products gives admissible limits with exactly the same initial datum. A vanishing seed and a cap hit alone do not imply that these limits differ: the hit might carry arbitrarily little mass. Excluding that loss of information is the central step.

1.3 From a cap hit to separation

Along a sequence of observation horizons shrinking to zero, rescale time and space at a cap time $t_{*,N}$ by $r_N = t_{*,N} + s_N$, and divide the hot density by r_N . The resulting cap is of order one. The cold density has a different normalization: multiplying it by r_N produces a locally bounded velocity measure at positive times. Its two axial velocity signs coexist in the limit because the increasingly fine spatial stripes average to their occupied fractions. Strong convergence of hot velocity averages identifies the cold-hot product. This use of velocity averaging follows the uniformly integrable formulation of [8, Proposition 3]. The limit obeys a linear Boltzmann equation in that two-sign background. This dilation is distinct from the fixed-horizon compactness used to construct admissible solutions.

To show that the dilated hot limit is nonzero, we analyze normalized Carleman gain kernels. Carleman's representation expresses a collision gain as an integral over velocity planes; its normalization here is derived in Section 3. Iterated-gain regularization has an earlier role in the homogeneous theory [12, Lemma 2.1]. In the present construction the estimates must be uniform as the cold velocity distributions collapse onto the axis, despite the singularity of H . Two complementary arguments provide this control. Near the velocity axis, two successive gains have vanishing normalized mass: the first has a uniformly non-atomic axial input, while a second large gain would force that input to match a small collection of characteristic cold speeds. Away from the axis, five gains suffice. Their times and angular parameters map to final time, position, and velocity with rank seven almost everywhere. A compact submersion argument then turns vanishing local hot mass into a vanishing gain contribution. Together these facts prevent the cap hit from disappearing in the weak limit.

The limiting hot equation also has a branching interpretation. A hot particle colliding with a cold jet creates two hot outputs. We retain finite collision histories with bounded velocities and show that their mass grows by a fixed factor inside an expanding spatial box. If the dilated observation horizon were infinite, this growth would exceed the polynomial mass bound supplied by the cap. The horizon is therefore finite, so the cap scale remains comparable to the physical observation horizon along the extracted sequence.

Finally, the dormant solution vanishes off the velocity axis under the corresponding shrinking-horizon dilation. If cap limits agreed with it at arbitrarily small fixed horizons, a diagonal choice of approximants would force the nonzero dilated hot limit to vanish as well. This contradiction yields two distinct early solutions of the original equation with the same datum. The periodic existence theorem then continues each solution globally while preserving admissibility. Figure 1 separates the two compactness procedures and their roles in this final comparison.

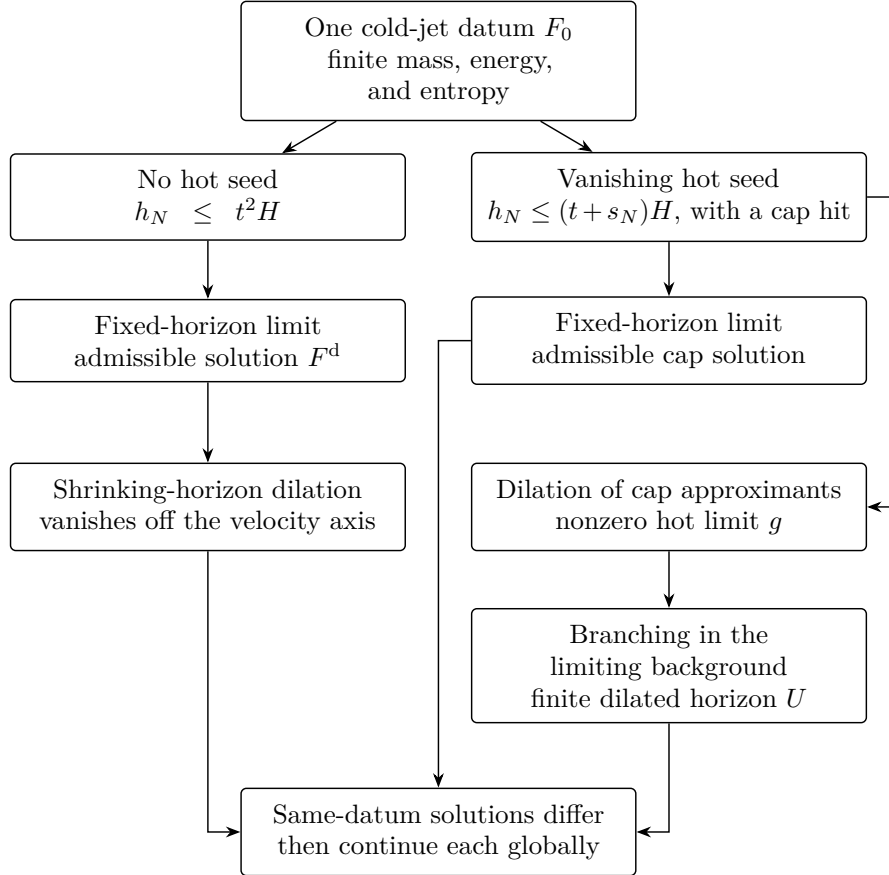


Figure 1: Proof structure. At each fixed physical horizon, the two approximation families yield admissible solutions with the same initial datum. Separately, cap approximants on shrinking horizons are dilated about a cap hit. Nonconcentration preserves a nonzero hot limit, and branching prevents its scale from vanishing relative to the horizon along the extracted sequence. These facts contradict equality with the dormant limit, whose corresponding dilation vanishes off the velocity axis.

1.4 Organization

Section 2 gives the full admissibility definition and the colored equations. Section 3 establishes uniform collision-kernel estimates, and Section 4 constructs the datum and controls its cold components. Section 5 selects the approximation families; Section 6 constructs their fixed-horizon limits and verifies admissibility. Section 7 identifies the dilated linear equation, and Section 8 proves that its hot limit is nonzero. Section 9 excludes an infinite dilated horizon. Section 10 completes the same-datum comparison and global continuation.

2 Equation, admissibility, and notation

The spatial torus $\mathbb{T}^3 = \mathbb{R}^3/\mathbb{Z}^3$ has volume one, and $\Omega = \mathbb{T}_x^3 \times \mathbb{R}_v^3$. We write $x = (y, z)$, with two transverse coordinates y and one axial coordinate z , and $v = (\bar{v}, v_3)$. In a periodic lift these coordinates range in $\mathbb{R}^2 \times \mathbb{R}$. All spatially rescaled limits below are taken in the lift. We write $D_t = \partial_t + v \cdot \nabla_x$ for free transport.

For incoming velocities v, v_* and $\omega \in S^2$, set

$$a = (v - v_*) \cdot \omega, \quad v' = v - a\omega, \quad v'_* = v_* + a\omega.$$

The surface measure on S^2 is its ordinary area measure. We use

$$\begin{aligned} Q(F, F) &= Q^+(F, F) - Q^-(F, F), \\ Q^+(F, F)(v) &= \int_{\mathbb{R}^3 \times S^2} |a| F(v') F(v'_*) dv_* d\omega, \\ Q^-(F, F)(v) &= F(v) \nu_F(v). \end{aligned} \tag{2.1}$$

Arguments t, x , identical in every factor, are suppressed. The associated symmetric gain is denoted $P(f, g)$, so that $P(F, F) = Q^+(F, F)$. Its symmetry in the hard-sphere normalization is included in Lemma 3.1. The loss frequency is

$$\nu_f(v) = 2\pi \int_{\mathbb{R}^3} |v - w| f(w) dw. \tag{2.2}$$

A superscript b on P, ν, Q means that the same factor $\mathbf{1}_{\{|v-v_*|<b\}}$ is inserted into gain and loss. This cutoff is invariant under elastic collision.

Put

$$M(v) = (2\pi)^{-3/2} e^{-|v|^2/2}, \quad G = F/M,$$

and define

$$\mathcal{H}_M(F) = \int_{\Omega} (F \log(F/M) - F + M) dx dv.$$

The entropy dissipation is

$$\mathcal{D}(F) = \frac{1}{4} \int_{\mathbb{T}^3 \times \mathbb{R}^3 \times \mathbb{R}^3 \times S^2} |a| (F' F'_* - F F_*) \log \frac{F' F'_*}{F F_*} dx dv dv_* d\omega.$$

The nonnegative integrand is assigned its lower-semicontinuous extended value when a factor vanishes. We use $0|\log 0| = 0$.

Definition 2.1 (Entropy-class solution). For a nonnegative datum F_0 satisfying

$$\int_{\Omega} F_0(1 + |v|^2 + |\log F_0|) dx dv < \infty,$$

an admissible global solution is a nonnegative F with the following properties.

- (i) $F \in C_w([0, \infty); L^1(\Omega))$, with $F(0) = F_0$, where weak continuity means continuity against every bounded measurable test. On each bounded time interval, $\int_{\Omega} F(1 + |v|^2 + |\log F|)$ is uniformly bounded.
- (ii) $Q^{\pm}(F, F)/(1 + G)$ belong to $L^1([0, T] \times \Omega)$ for every finite T . For every $\beta \in C^1([0, \infty))$ with $\beta(0) = 0$ and $|\beta'(z)| \leq C_{\beta}/(1 + z)$, and every smooth test $\varphi(t, x, v)$ periodic in x and compactly supported in $[0, \infty) \times \mathbb{R}_v^3$,

$$\begin{aligned} \int_0^{\infty} \int_{\Omega} [M\beta(G)(\partial_t \varphi + v \cdot \nabla_x \varphi) + \beta'(G)Q(F, F)\varphi] dx dv dt \\ + \int_{\Omega} M\beta(F_0/M)\varphi(0, x, v) dx dv = 0. \end{aligned}$$

- (iii) With $\rho = \int F dv$ and $j = \int vF dv$, $\partial_t \rho + \nabla_x \cdot j = 0$ distributionally with the initial density $\int F_0 dv$. Total momentum is conserved. For every $t \geq 0$,

$$\int_{\Omega} |v|^2 F(t) \leq \int_{\Omega} |v|^2 F_0, \quad \mathcal{H}_M(F(t)) + \int_0^t \mathcal{D}(F(s)) ds \leq \mathcal{H}_M(F_0).$$

The same definition on a finite interval uses tests and traces on that interval. We use the global existence theory [3, 4, 11] only to continue solutions already constructed on a short interval. In particular, we do not require local momentum or local energy conservation as additional admissibility conditions.

2.1 A velocity weight and colored components

Fix $1 < q < 2$ and $m = 100$. The weight used throughout is

$$H(v) = e^{-|v|^2} \langle v \rangle^{-m} (1 + |\bar{v}|^{-q}), \quad \langle v \rangle = (1 + |v|^2)^{1/2}. \quad (2.3)$$

Its values on the velocity axis can be assigned arbitrarily; all weighted supremum estimates are essential estimates. It is integrable with every polynomial velocity weight and lies in L^p for some $p > 1$. Write $\|f\|_H = \text{ess sup}_{x,v} |f(x, v)|/H(v)$.

The cold components are c_i , their sum is c , and h denotes the hot component. Particles retain a cold label only in collisions within that label. All other collisions create hot particles. With $F = c + h$, this gives

$$\begin{aligned} (D_t)c_i &= P(c_i, c_i) - c_i \nu_F, \\ (D_t)h &= 2P(c, h) + P(h, h) + \sum_{i \neq j} P(c_i, c_j) - h \nu_F. \end{aligned} \quad (2.4)$$

The sum over $i \neq j$ is ordered. Summing these equations gives $(D_t)F = Q(F, F)$. This identity also holds with a common collision cutoff. When labels include periodic copies, the individual functions are defined in the lift and are covariant under translations; their sums are periodic.

Constants denoted C , with or without subscripts, may depend on the fixed construction parameters. They never depend on the approximation index or the shrinking time horizon unless explicitly stated. The parameters needed for branching growth will be fixed before the final small geometric and time parameters. Section 9 records the order of choices.

3 Collision kernels and a velocity majorant

The construction in the next section uses cold velocity distributions with arbitrarily small transverse widths. We seek bounds for $P(c, H)/H$ that remain uniform as those widths tend to zero, together with a bound for $P(H, H)/H$. The later nonconcentration argument needs more than bounded total gain: it also needs control of the complete normalized measure in the cold and hot input velocities. We obtain both kinds of estimates below.

Write $v = (\bar{v}, v_3) \in \mathbb{R}^2 \times \mathbb{R}$ and set

$$W(v) = 1 + |\bar{v}|^{-q}, \quad H(v) = e^{-|v|^2} \langle v \rangle^{-m} W(v), \quad 1 < q < 2, \quad m = 100,$$

as in (2.3). Values on the velocity axis are irrelevant to all the almost-everywhere estimates below. The function H has every polynomial moment and belongs to $L^p(\mathbb{R}^3)$ whenever $1 \leq p < 2/q$. Indeed, its only local singularity is $|\bar{v}|^{-q}$ in two dimensions, and its decay at infinity is Gaussian.

3.1 The exact Carleman formula

The hyperplane representation used here is a Carleman formula; see also [6, Appendix C]. We derive its coefficient for the normalization of this paper.

Lemma 3.1 (Collision geometry). *For the normalization in (2.1), the symmetric bilinear gain is*

$$P(c, h)(v) = 2 \int_{\mathbb{R}^3} c(p) \, dp \int_{\mathbb{R}^3} \delta((v - p) \cdot (\xi - v)) h(\xi) \, d\xi. \quad (3.1)$$

For fixed $p \neq v$, the inner delta denotes surface measure on the plane $(v - p) \cdot (\xi - v) = 0$, divided by $|v - p|$. For incoming velocities p, ξ , the outgoing measure has total mass $2\pi|p - \xi|$ and is uniform on the sphere with diameter $[p, \xi]$. Formula (3.1) also defines the gain when c is a nonnegative velocity measure and h is a nonnegative density, whenever the displayed integral is finite almost everywhere.

Proof. Put $g = p - \xi$. Reflection sends g to $g - 2(g \cdot \omega)\omega$. On either hemisphere of the normal sphere, the change from the polar angle of ω to the reflected direction shows that $|g \cdot \omega| \, d\omega$ pushes forward to $(|g|/4) \, d\sigma$. The two hemispheres therefore give $(|g|/2) \, d\sigma$, where $\sigma \in S^2$ is the outgoing relative direction. Equivalently, for every nonnegative test function ψ , the one-output measure is

$$\frac{|p - \xi|}{2} \int_{S^2} \psi \left(\frac{p + \xi}{2} + \frac{|p - \xi|}{2} \sigma \right) \, d\sigma. \quad (3.2)$$

Its mass is $2\pi|p - \xi|$. The affine map in this integral multiplies surface area by $|p - \xi|^2/4$, so the density on its image sphere is $2/|p - \xi|$.

The same sphere is the zero set of $f(v) = (v - p) \cdot (\xi - v)$. On that sphere, $|\nabla_v f(v)| = |p - \xi|$. The coarea formula consequently identifies the measure in (3.2) with $2\delta(f(v)) \, dv$. Integration against the two incoming densities and Tonelli's theorem give (3.1). The usual pre/post-collisional change of variables identifies this incoming-pair expression with the gain in (2.1). Coincident incoming velocities have zero collision rate. The same identities can first be integrated against a velocity measure $c(dp)$, proving the asserted extension. $\square$

The cutoff gain obtained by inserting $\mathbf{1}_{|p - \xi| < b}$ is bounded by the uncut gain. Thus all upper estimates in this section also hold for the collision cutoffs used later.

3.2 A geometric slice estimate

For $d > 0$, let $\mathcal{G}_d$ be the centered transverse Gaussian probability measure with density

$$\frac{1}{\pi d^2} e^{-|\bar{p}|^2/d^2},$$

and let $\mathcal{G}_0 = \delta_0$. We shall also use the uniform probability measure $\mathcal{U}_d$ on $\{\bar{p} : |\bar{p}| < d\}$.

Lemma 3.2 (Transverse averaging and axial slices). *Let λ be a probability mixture of the measures $\mathcal{U}_d$, $d > 0$, and δ_0 , and let ρ be any axial probability measure. No bound on its density or its support is required. Put $c = \lambda \otimes \rho$. For $R \geq 1$, $\bar{v} \neq 0$, and an interval $I \subset \mathbb{R}$ of length $0 < \ell \leq 2R$, one has*

$$\frac{1}{W(v)} \int c(dp) \int_{|\xi-v| \leq R} \delta((v-p) \cdot (\xi-v)) W(\xi) d\xi \leq C_q R^2, \quad (3.3)$$

$$\frac{1}{W(v)} \int c(dp) \int_{\substack{|\xi-v| \leq R \\ \xi_3 \in I}} \delta((v-p) \cdot (\xi-v)) W(\xi) d\xi \leq C_q (R\ell + R^{q-1} \ell^{2-q}). \quad (3.4)$$

The constants are independent of all transverse widths and of ρ . In particular, these statements hold with $\lambda = \mathcal{G}_d$ for every $d \geq 0$.

Proof. Set $r = |\bar{v}|$, $n = \bar{v} - \bar{p}$, and $e_n = n/|n|$. Except on a set of transverse measure zero, $n \neq 0$. For fixed p_3, ξ_3 , the delta in the transverse variables is line measure divided by $|n|$, on a line whose signed distance from zero is

$$A - B(\xi_3 - v_3), \quad A = e_n \cdot \bar{v}, \quad B = \frac{v_3 - p_3}{|n|}.$$

The regular part of its integral over $|\xi - v| \leq R$ is at most $2R/|n|$. The singular part is bounded by its integral over the full line, namely

$$\frac{1}{|n|} \int_{\mathbb{R}} ((A - B(\xi_3 - v_3))^2 + s^2)^{-q/2} ds = \frac{a_q}{|n|} |A - B(\xi_3 - v_3)|^{1-q}, \quad (3.5)$$

where $a_q = \int_{\mathbb{R}} (1 + s^2)^{-q/2} ds < \infty$ because $q > 1$. Infinite values on a zero-distance slice are harmless in the nonnegative iterated integral.

For every interval $J \subset [-R, R]$ of length at most ℓ ,

$$\int_J |A - Bx|^{1-q} dx \leq C_q R^{q-1} \ell^{2-q} |A|^{1-q}. \quad (3.6)$$

If $2R|B| \leq |A|$, the integrand is at most $C_q |A|^{1-q}$, and the assertion follows from $\ell \leq 2R$. Otherwise, integration about the possible root gives an upper bound $C_q |B|^{1-q} \ell^{2-q}$; since $1 - q < 0$ and $|B| > |A|/(2R)$, this proves (3.6). When $A = 0$ the displayed upper bound is understood in the extended sense.

We next prove

$$\int \frac{1 + |e_n \cdot e_{\bar{v}}|^{1-q}}{|n|} \lambda(d\bar{p}) \leq \frac{C_q}{r}. \quad (3.7)$$

It suffices to consider a disk of radius d . If $d \leq r/2$, then $|n| \geq r/2$ and $e_n \cdot e_{\bar{v}} \geq 1/3$, which proves the bound. If $d > r/2$, use polar coordinates for n . Its range lies in $|n| < r + d < 3d$. The factor $|n|^{-1}$ cancels the radial Jacobian, while

$$\int_0^{2\pi} (1 + |\cos \theta|^{1-q}) d\theta < \infty$$

because $q < 2$. Division by the disk area gives $C_q/d \leq 2C_q/r$. The point mass at zero satisfies the same estimate directly. Finally, the Gaussian is a probability mixture of these disks: for $d > 0$ its mixture density in the disk radius a is

$$\frac{2a^3}{d^4} e^{-a^2/d^2} da,$$

whose integral is one. This proves the Gaussian assertion as well.

Integrating (3.5) in ξ_3 , applying (3.6), and then (3.7), bounds the expression in (3.4) by

$$C_q R \ell \frac{r^{-1}}{1 + r^{-q}} + C_q R^{q-1} \ell^{2-q} \frac{r^{-q}}{1 + r^{-q}}.$$

Both fractions are bounded for $q > 1$. These bounds are independent of p_3 , so integration against the arbitrary probability ρ preserves them. Taking the entire axial range of length $2R$ also gives (3.3). $\square$

3.3 Sheets and normalized kernel measures

Definition 3.3 (Uniform sheet class). Fix $L \geq 1$. An elementary sheet in $\mathcal{S}_L$ is a product probability measure

$$c(dp) = \mathcal{G}_d(d\bar{p})\rho(dp_3), \quad 0 \leq d \leq \frac{1}{8}, \quad \int_{\mathbb{R}} e^{8a^2} \rho(da) \leq L. \quad (3.8)$$

We also allow probability mixtures of elementary sheets. An upper bound by a sheet with amplitude A_0 means domination as measures by A_0 times such a mixture.

Every axial probability supported in $[-A, A]$ satisfies this definition with $L = e^{8A^2}$. In particular, axial atoms and limits of arbitrarily narrow axial densities are permitted. The exponential moment is used only to remove large axial velocities, and imposes no axial regularity.

For such a sheet and $\bar{v} \neq 0$, define its normalized kernel by

$$\mathcal{K}_c(v; dp d\xi) = 2c(dp) \delta((v-p) \cdot (\xi-v)) \frac{H(\xi)}{H(v)} d\xi. \quad (3.9)$$

The total mass is $P(c, H)(v)/H(v)$. Kernel restrictions always refer to this complete measure, including the ratio $H(\xi)/H(v)$.

Lemma 3.4 (Uniform sheet gain bound). *For every $L < \infty$ there is $C_{q,m,L} < \infty$ such that*

$$P(c, H)(v) \leq C_{q,m,L} H(v) \quad \text{for almost every } v, \quad c \in \mathcal{S}_L. \quad (3.10)$$

More quantitatively, for $R \geq 1$,

$$\mathcal{K}_c(v; R < |p| + |\xi - v| \leq 2R) \leq C_{q,m,L} R^2 e^{-cR^2}, \quad (3.11)$$

where $c > 0$ is an absolute constant. The same conclusions, multiplied by A_0 , hold under sheet domination with amplitude A_0 .

Proof. By positivity it suffices to treat an elementary sheet. Put $k = \xi - v$ and $w = p + k$. On the collision plane,

$$|p|^2 + |\xi|^2 = |v|^2 + |w|^2, \quad \frac{H(\xi)}{H(v)} = e^{|p|^2 - |w|^2} \frac{\langle v \rangle^m W(\xi)}{\langle \xi \rangle^m W(v)}. \quad (3.12)$$

The same identity implies $\langle v \rangle \leq \langle p \rangle \langle \xi \rangle$. Since $\langle p \rangle^m \leq C_m e^{|p|^2}$, the kernel is bounded by

$$C_m e^{2|p|^2} c(dp) e^{-|w|^2} \delta((v - p) \cdot k) \frac{W(v + k)}{W(v)} dk. \quad (3.13)$$

The tilted cold measure retains a product majorant. For $0 < d \leq 1/8$,

$$e^{2|\bar{p}|^2} \mathcal{G}_d(d\bar{p}) \leq 4e^{-46|\bar{p}|^2} \mathcal{G}_{2d}(d\bar{p}). \quad (3.14)$$

This follows by dividing the two Gaussian densities: their ratio is $4 \exp[-(3/(4d^2) - 2)|\bar{p}|^2]$, and $3/(4d^2) - 2 \geq 46$. The assertion at $d = 0$ has the evident interpretation. The axial tilted measure $\tilde{\rho}(da) = e^{2a^2} \rho(da)$ has mass at most L , and satisfies

$$\tilde{\rho}\{|a| > u\} \leq L e^{-6u^2}, \quad u \geq 0. \quad (3.15)$$

Consequently $e^{2|p|^2} c(dp)$ is bounded by $4L$ times a product of a centered transverse Gaussian and an axial probability. Moreover, its restriction to $|p| > u$ is bounded by CLe^{-cu^2} times a sum of at most two such product probabilities. To see the last assertion, split into $|\bar{p}| > u/\sqrt{2}$ and $|p_3| > u/\sqrt{2}$, and apply (3.14) and (3.15), respectively. The axial probabilities obtained by normalization may have large support; Lemma 3.2 permits this.

On the shell in (3.11), either $|p| > R/4$ or $|w| > R/2$, because $|p| + |k| \leq 2|p| + |w|$. In the first case the preceding tilted-measure tail contributes CLe^{-cR^2} . In the second case $e^{-|w|^2} \leq e^{-R^2/4}$. In either case the remaining integral has $|k| \leq 2R$ and is bounded by $C_q R^2$ using Lemma 3.2. This proves (3.11). The base region $|p| + |k| \leq 2$ is bounded by the same geometric lemma and the total tilted mass. Summing the dyadic shells proves (3.10) and also the finiteness of (3.9). $\square$

The sheet gain bound controls total normalized mass. We next control its velocity tails and its behavior as the outgoing velocity approaches the axis. The key distinction is that the hot input can concentrate transversely on the axis while its axial marginal remains uniformly non-atomic.

Lemma 3.5 (Uniform tightness and axis estimates). *Fix $L < \infty$. The following conclusions hold uniformly for $c \in \mathcal{S}_L$ and for outgoing velocities off the axis.*

(i) *The tails in $|p| + |\xi - v|$ are uniformly negligible:*

$$\lim_{R \rightarrow \infty} \sup_{c \in \mathcal{S}_L} \operatorname{ess\,sup}_v \mathcal{K}_c(v; |p| + |\xi - v| > R) = 0. \quad (3.16)$$

Also,

$$\lim_{M \rightarrow \infty} \sup_{c \in \mathcal{S}_L} \operatorname{ess\,sup}_{|v| > M} \mathcal{K}_c(v; \mathbb{R}^3 \times \mathbb{R}^3) = 0. \quad (3.17)$$

(ii) For every fixed $M, \delta > 0$,

$$\lim_{\varepsilon \downarrow 0} \sup_{c \in \mathcal{S}_L} \operatorname{ess\,sup}_{\substack{|v| \leq M \\ 0 < |\bar{v}| \leq \varepsilon}} \mathcal{K}_c(v; |\bar{\xi}| \geq \delta) = 0, \quad (3.18)$$

$$\lim_{\varepsilon \downarrow 0} \sup_{c \in \mathcal{S}_L} \operatorname{ess\,sup}_{\substack{|v| \leq M \\ 0 < |\bar{v}| \leq \varepsilon}} \mathcal{K}_c(v; |v_3 - p_3| \geq \delta) = 0. \quad (3.19)$$

(iii) The axial input marginals are uniformly non-atomic. More precisely, for each $R \geq 1$ and each interval I of length $0 < \ell \leq 1$,

$$\mathcal{K}_c(v; |p| + |\xi - v| \leq R, \xi_3 \in I) \leq C_{q,m,L,R}(\ell + \ell^{2-q}). \quad (3.20)$$

In combination with (3.16), this gives

$$\lim_{\ell \downarrow 0} \sup_{c \in \mathcal{S}_L} \operatorname{ess\,sup}_v \sup_{|I| \leq \ell} \mathcal{K}_c(v; \xi_3 \in I) = 0. \quad (3.21)$$

(iv) Let $\mathcal{S}_L(\varepsilon)$ denote the subclass whose elementary transverse widths are at most ε . For every $M < \infty$ and $r_0 > 0$,

$$\lim_{\delta \downarrow 0} \limsup_{\varepsilon \downarrow 0} \sup_{c \in \mathcal{S}_L(\varepsilon)} \operatorname{ess\,sup}_{\substack{|v| \leq M \\ |\bar{v}| \geq r_0}} \mathcal{K}_c(v; |\bar{\xi}| < \delta) = 0. \quad (3.22)$$

All these conclusions hold under uniformly bounded sheet amplitudes.

Proof. The dyadic estimate (3.11) proves (3.16). We may consequently work on $|p| + |k| \leq R$ and remove that restriction at the end of each argument. On this region the exponential and polynomial factors in (3.12) are bounded by a constant depending on R .

For (3.17), if $|\bar{v}| > 2R + 1$, then $|\bar{v} - \bar{p}| \geq |\bar{v}|/2$ and $|\bar{\xi}| \geq |\bar{v}| - R$. The transverse line estimate, followed by the axial integration over a length at most $2R$, gives $C_R/|\bar{v}|$. If $|v_3| > 2R + 1$, integrate the delta in ξ_3 instead. Its divisor is $|v_3 - p_3| \geq |v_3|/2$, and

$$\sup_{a \in \mathbb{R}^2} \int_{|\zeta - a| \leq R} (1 + |\zeta|^{-q}) \, d\zeta \leq C_q(R^2 + R^{2-q}). \quad (3.23)$$

For completeness, if $|a| \leq 2R$ the disk is contained in the disk of radius $3R$ about zero; if $|a| > 2R$, then $|\zeta| > |a|/2 > R$ on the disk, giving the same bound. We obtain $C_R/|v_3|$. At least one of the two outgoing components becomes large with $|v|$, proving (3.17).

For (3.18), on $|\bar{\xi}| \geq \delta$ the factor $W(\xi)$ is bounded by $1 + \delta^{-q}$. The regular line estimate and (3.7) therefore give

$$C_{R,\delta} \frac{|\bar{v}|^{-1}}{1 + |\bar{v}|^{-q}} \leq C_{R,\delta} |\bar{v}|^{q-1} \quad (0 < |\bar{v}| \leq 1).$$

For (3.19), integrate in ξ_3 first, using $|v_3 - p_3|^{-1} \leq \delta^{-1}$ and (3.23). The bound is $C_{R,\delta}/W(v) \leq C_{R,\delta}|\bar{v}|^q$. These are uniform in the axial law. For (3.20), discard the bounded exponential and polynomial factors and apply (3.4). This also proves (3.21) after the tail removal.

It remains to prove (3.22). Fix the truncation radius R . The part $|\bar{p}| > r_0/4$ is uniformly negligible as the transverse widths tend to zero. Indeed,

$$\mathbf{1}_{|\bar{p}| > r_0/4} \mathcal{G}_d(d\bar{p}) \leq 2e^{-r_0^2/(32d^2)} \mathcal{G}_{\sqrt{2}d}(d\bar{p}),$$

and Lemma 3.4 applies to the wider Gaussian for all sufficiently small d . Now retain $|\bar{p}| \leq r_0/4$. With the notation of Lemma 3.2,

$$|n| \geq \frac{3}{4}r_0, \quad A = \frac{(\bar{v} - \bar{p}) \cdot \bar{v}}{|\bar{v} - \bar{p}|} \geq \frac{3}{5}r_0.$$

If $|v_3 - p_3| \leq r_0^2/(8R)$, then for $|\xi_3 - v_3| \leq R$,

$$|B(\xi_3 - v_3)| \leq \frac{1}{6}r_0.$$

The transverse collision line stays at distance greater than $r_0/3$ from zero, and cannot meet $|\bar{\xi}| < \delta$ when $\delta < r_0/3$. In the complementary case, $|v_3 - p_3|^{-1} \leq 8R/r_0^2$. Integrating first in ξ_3 bounds the small-input contribution by

$$C_{R,r_0} \int_{|\bar{\xi}| < \delta} (1 + |\bar{\xi}|^{-q}) d\bar{\xi} \leq C_{R,r_0} (\delta^2 + \delta^{2-q}).$$

First let the widths tend to zero, then δ tend to zero, and finally remove the truncation. This proves the last assertion. $\square$

3.4 Gaussian parameter laws and compressed axial maps

The next formulation records the uniformity needed when a cold profile is parameterized before its axial change of variables.

Lemma 3.6 (Gaussian parameter tails). *Let a measure in parameters $w = (\bar{w}, w_3) \in \mathbb{R}^2 \times \mathbb{R}$ have density j satisfying*

$$0 \leq j(w) \leq C_0 \pi^{-3/2} e^{-|w|^2}.$$

Let its velocity image be given by

$$\bar{p} = d\bar{w}, \quad p_3 = \Phi(w_3), \quad 0 \leq d \leq \frac{1}{16}, \quad |\Phi(a)| \leq A + \delta|a|, \quad 0 \leq \delta \leq \frac{1}{16}, \quad (3.24)$$

where Φ is merely measurable. The image measure is dominated by C_0 times a sheet in $\mathcal{S}_{L_A}$, with L_A depending only on A . Its gain satisfies Lemma 3.4 with a constant depending only on q, m, A, C_0 .

If $c_{>B}$ is the image of the restricted parameter measure $\mathbf{1}_{|w|>B} j(w) dw$, then

$$\text{ess sup}_v \frac{P(c_{>B}, H)(v)}{H(v)} \leq C_{q,m,A,C_0} e^{-B^2/2}, \quad B \geq 0. \quad (3.25)$$

Thus parameter truncations are valid in the normalized collision measure, even when Φ compresses intervals or has atoms in its image law. Probability mixtures of measurable families with common q, m, A, C_0 and the width bounds in (3.24) preserve these assertions. More generally, a positive mixing measure of total mass $B_0 < \infty$ multiplies the domination amplitude and quantitative upper bounds by B_0 ; the sheet parameter L_A is unchanged. For (3.25), impose the restriction $|w| > B$ in each component before mixing. Additional measurable masks taking values in $[0, 1]$ preserve the assertions.

Proof. The Gaussian majorant is a product measure. Its transverse image is $\mathcal{G}_d$, and its axial image is the probability $\rho = \Phi_*(\pi^{-1/2}e^{-a^2} da)$. There is no need to differentiate or invert Φ . From (3.24),

$$e^{8|\Phi(a)|^2} \leq e^{16A^2} e^{16\delta^2 a^2}.$$

Even under the wider axial probability $(2\pi)^{-1/2}e^{-a^2/2} da$, the integral of the right side is

$$e^{16A^2} (1 - 32\delta^2)^{-1/2} \leq 2e^{16A^2}.$$

We may therefore take $L_A = 2e^{16A^2}$ for both the original and the wider parameter laws. The original image has the required sheet domination.

On the parameter tail,

$$\mathbf{1}_{|w|>B} e^{-|w|^2} \leq e^{-B^2/2} e^{-|w|^2/2}.$$

After normalization the right side is $2^{3/2}e^{-B^2/2}$ times a Gaussian probability with parameter widths enlarged by $\sqrt{2}$. Its transverse velocity width is at most $\sqrt{2}/16 < 1/8$, and its axial image has the exponential moment just proved. Lemma 3.4 applied to this dominating image gives (3.25).

For a probability mixture, mix the dominating sheets just constructed; this is again a member of $\mathcal{S}_{L_A}$. If the mixing measure has mass $B_0 > 0$, normalize it by B_0 and retain that factor in the domination amplitude. If $B_0 = 0$, the mixed image is zero. The same argument applied to the wider Gaussian majorants of the component tails proves the asserted factor in (3.25). Thus the estimates are uniform over families of mixtures whose masses B_0 have a common finite bound independent of the tail cutoff. Masks taking values in $[0, 1]$ only decrease the measures, whether imposed before or after the pushforward. $\square$

For example, suppose an axial velocity is determined by $p_3 = a(z - tp_3) + dw_3$, where a is nondecreasing, and u solves $u = a(z - tu)$. The function $p \mapsto p - a(z - tp)$ is increasing with inverse Lipschitz constant at most one. Hence

$$|p_3 - u| \leq d|w_3|. \quad (3.26)$$

If $|u| \leq A$, Lemma 3.6 applies. A large Jacobian compression of the axial law does not affect its constants. Gaussian majorants with any fixed positive exponent in place of one are covered by a fixed rescaling of the parameters and a corresponding decrease of the upper bounds on the physical widths.

3.5 The hot-hot gain

Lemma 3.7 (Quadratic gain bound). *There is a constant $C_{q,m}$ such that*

$$P(H, H) \leq C_{q,m} H \quad \text{almost everywhere.} \quad (3.27)$$

In fact the proof below works for every $m > 5$.

Proof. The integrand is symmetric in p, ξ . On the collision plane, $|v|^2 \leq |p|^2 + |\xi|^2$, so at least one incoming speed is at least $|v|/2$. It suffices, at a factor of two, to retain $|\xi| \geq |v|/2$. Formula (3.12) now gives

$$\frac{P(H, H)(v)}{H(v)} \leq C_m \int_{|\xi| \geq |v|/2} \delta((v - p) \cdot (\xi - v)) e^{-|p+\xi-v|^2} \langle p \rangle^{-m} \frac{W(p)W(\xi)}{W(v)} dp d\xi, \quad (3.28)$$

because $\langle v \rangle^m / \langle \xi \rangle^m \leq 2^m$.

Again write $k = \xi - v$ and $w = p + k$. On $R < |p| + |k| \leq 2R$, $R \geq 1$, either $|p| > R/4$ or $|w| > R/2$. Consequently

$$\langle p \rangle^{-m} e^{-|w|^2} \leq C_m R^{-m} e^{-|w|^2/2}. \quad (3.29)$$

In the second alternative this uses the finite supremum of $s^m e^{-s^2/2}$; in the first it follows directly from the polynomial factor.

The measure $W(p) \mathbf{1}_{|p| \leq 2R} dp$ is dominated by

$$(1 + |\bar{p}|^{-q}) \mathbf{1}_{|\bar{p}| \leq 2R} d\bar{p} \otimes \mathbf{1}_{|p_3| \leq 2R} dp_3.$$

Its transverse factor is a positive mixture of centered disks. Indeed, for $D > 0$,

$$|\bar{p}|^{-q} \mathbf{1}_{|\bar{p}| \leq D} = D^{-q} \mathbf{1}_{|\bar{p}| \leq D} + q \int_0^D a^{-q-1} \mathbf{1}_{|\bar{p}| \leq a} da$$

almost everywhere. Upon normalizing the disks, the mixture has total mass $\pi D^2 + 2\pi D^{2-q}/(2-q)$. The axial factor has mass $4R$. For $D = 2R$ the resulting product mixture therefore has total mass at most $C_q R^3$.

Discard $e^{-|w|^2/2}$ and the incoming-speed restriction in (3.28). Apply Lemma 3.2 to each normalized product in this mixture, with $|k| \leq 2R$. In conjunction with (3.29), this yields the shell estimate

$$\left. \frac{P(H, H)(v)}{H(v)} \right|_{R < |p| + |\xi - v| \leq 2R} \leq C_{q,m} R^{5-m}. \quad (3.30)$$

The notation on the left denotes the restricted nonnegative integral from (3.28), including its fixed symmetry factor. The base region is bounded by the same disk-mixture argument with $R = 1$. Summation over dyadic R proves the result for $m > 5$. $\square$

Remark 3.8 (A sharp axial endpoint). A single normalized gain need not become small as the outgoing velocity approaches the axis. Take $c = \delta_{(0,0,u)}$ and $v_\varepsilon = (\varepsilon, 0, u)$. The collision plane is $\xi_1 = \varepsilon$, and

$$\frac{P(c, H)(v_\varepsilon)}{H(v_\varepsilon)} = \frac{2}{\varepsilon H(v_\varepsilon)} \int_{\mathbb{R}^2} H(\varepsilon, y, z) dy dz.$$

The regular part of $W(\varepsilon, y, z)$ contributes $O(\varepsilon^{q-1})$. In its singular part substitute $y = \varepsilon s$. Dominated convergence, with dominating function $(1 + s^2)^{-q/2} e^{-z^2} \langle z \rangle^{-m}$, gives

$$\lim_{\varepsilon \downarrow 0} \frac{P(c, H)(v_\varepsilon)}{H(v_\varepsilon)} = 2a_q e^{u^2} \langle u \rangle^m \int_{\mathbb{R}} e^{-z^2} \langle z \rangle^{-m} dz > 0.$$

The corresponding axial input marginal converges to the absolutely continuous measure

$$2a_q e^{u^2} \langle u \rangle^m e^{-z^2} \langle z \rangle^{-m} dz.$$

Thus transverse concentration is compatible with the uniform axial interval estimate. This is the reason for using two successive gains in the axial part of the later nonconcentration argument.

4 Cold jets and uniform color estimates

We construct the common finite-entropy datum and bound each cold color by its free transport plus a small Maxwellian remainder. Narrow velocity profiles control self-collisions, while spatial gaps make collisions between distinct colors small. These estimates supply the cold density bound and the mixed-cold source bound used to select the hot caps in Section 5.

The estimates in this section hold for every fixed choice

$$K \geq 1, \quad 0 < \eta \leq \log 2, \quad 0 < \alpha \leq \frac{1}{4}, \quad 0 < \gamma \leq 1.$$

These four parameters will subsequently be fixed by the growth argument. Here we choose the geometric and thermal parameters after them. To keep the later shortening of the time interval separate from the estimates, fix the preliminary bounds

$$\bar{s} = \frac{1}{8}, \quad \bar{T} = \frac{1}{32}, \quad 0 < s_0 \leq \bar{s}, \quad 0 < T_0 \leq \bar{T}. \quad (4.1)$$

All estimates proved below hold on the preliminary interval $[0, \bar{T}]$. Their constants do not change when T_0 is subsequently decreased. They may depend on the fixed parameters and on the fixed function J , but never on the number of retained annuli, the collision cutoff, or the size of the hot component.

4.1 The geometry and the datum

Put

$$s_j = s_0 e^{-j\eta}, \quad p_j = s_j^2, \quad c_\eta = \frac{e^\eta - 1}{16}, \quad \delta_j = \frac{c_\eta}{4} s_j^2, \quad j \geq 1.$$

For $s = s_j$, define the annular core and the two stripe patterns by

$$\begin{aligned} A_j &= \{y \in \mathbb{R}^2 : s_j + c_\eta p_j < |y| < e^\eta s_j - c_\eta p_j\}, \\ P_j^+ &= \bigcup_{k \in \mathbb{Z}} \{y : kp_j < y_1 < (k + \tfrac{1}{4})p_j\}, \\ P_j^- &= \bigcup_{k \in \mathbb{Z}} \{y : (k + \tfrac{1}{2})p_j < y_1 < (k + \tfrac{3}{4})p_j\}, \\ E_j^\sigma &= A_j \cap P_j^\sigma, \quad \sigma \in \{-1, +1\}. \end{aligned}$$

Each stripe pattern has period p_j and fraction $\theta = 1/4$. This is a statement about the periodic pattern: intersecting it with the annulus need not preserve the fraction exactly. We retain the original sets E_j^σ in the datum.

Let

$$\hat{E}_j^\sigma = \{y : \text{dist}(y, E_j^\sigma) \leq \delta_j\}, \quad a = \frac{1}{8}, \quad b_{\text{sp}} = \frac{1}{16}.$$

For a label $i = (j, \sigma, n)$, $n = (\bar{n}, n_3) \in \mathbb{Z}^3$, set

$$\mathcal{O}_i(t) = (\hat{E}_j^\sigma + \bar{n}) \times [n_3 - a - 2t - b_{\text{sp}}, n_3 + a + 2t + b_{\text{sp}}]. \quad (4.2)$$

These enlargements will only localize upper bounds; they will not replace the stripe masks when their average is computed.

Lemma 4.1 (Disjoint enlargements). *For $0 \leq t \leq \bar{T}$, the sets $\mathcal{O}_i(t)$ are pairwise disjoint. Moreover,*

$$\widehat{E}_j^\sigma \subset \{s_j < |y| < e^\eta s_j\} \subset B(0, s_0).$$

At each point of the periodic lift there is therefore at most one label whose enlargement contains that point.

Proof. Opposite stripe patterns at the same scale have distance at least $p_j/4$, whereas $2\delta_j = c_\eta p_j/2 < p_j/4$. The radial gap between the cores for the adjacent scales s_j and s_{j+1} is

$$c_\eta(p_j + p_{j+1}) > \delta_j + \delta_{j+1}.$$

The gaps between nonadjacent cores are larger. Enlarging by $\delta_j < c_\eta p_j$ still leaves the j -th set inside its original open annulus. This proves disjointness of all the untranslated transverse enlargements. They lie in $B(0, s_0)$, so their distinct integer translates are disjoint as well. Finally,

$$a + 2\bar{T} + b_{\text{sp}} = \frac{1}{4}.$$

The axial intervals belonging to distinct integers n_3 are disjoint. These two observations prove the assertion for the products in (4.2). $\square$

Fix a nonnegative smooth probability density J , supported in the unit ball of $\mathbb{R}^3$. Write

$$S(\ell) = \text{sech}^2 \ell, \quad a_s^\sigma(z) = \sigma + \alpha \tanh(\gamma z/s), \quad d_s = d_0 s^{10}.$$

The positive parameter $d_0 \leq 1$ will be decreased below finitely many times. In particular, we require that

$$d_0 \bar{s}^{10} \leq \frac{1}{64}, \quad \bar{T} d_0 \bar{s}^8 \leq c_\eta/8. \quad (4.3)$$

The width $1/64$, including the fixed Gaussian widenings used below, lies within the range of Lemma 3.4. For $i = (j, \sigma, n)$, with $s = s_j$, define on the lift

$$c_i^0(x, v) = \mathbf{1}_{E_j^\sigma}(y - \bar{n}) \mathbf{1}_{(-a, a)}(z - n_3) \frac{K}{s} S\left(\frac{\gamma(z - n_3)}{s}\right) d_s^{-3} J\left(\frac{v - a_s^\sigma(z - n_3)e_3}{d_s}\right).$$

The periodic datum is

$$F_0(x, v) = \sum_{j \geq 1} \sum_{\sigma = \pm 1} \sum_{n \in \mathbb{Z}^3} c_{(j, \sigma, n)}^0(x, v). \quad (4.4)$$

At a given spatial point at most one summand is nonzero. Integer translations permute the labels, so the sum is periodic.

Lemma 4.2 (Integrability of the datum). *The datum (4.4) satisfies*

$$\int_{\Omega} F_0(1 + |v|^2 + |\log F_0|) \, dx \, dv < \infty.$$

Its truncations $F_0^{(N)}$, retaining $j \leq N$, converge to F_0 in mass, energy, and relative entropy. In addition, if $\chi \geq 0$ is a fixed bounded smooth compactly supported function of (x, v) , periodized after replacing x by x/s_N , and $0 \leq \lambda_N \leq L_0 < \infty$, then

$$F_0^{(N)} + \lambda_N s_N \chi(x/s_N, v)$$

has the same limiting mass, energy, and relative entropy as F_0 . For the last assertion s_0 may be decreased so that distinct periodic copies of the spatial support of the seed are disjoint.

Proof. Integration over one period and summation over integer copies equal integration of each unshifted prototype over $\mathbb{R}_x^3$. The transverse area of E_j^σ is at most $C_\eta s_j^2$, and

$$\int_{-a}^a S(\gamma z/s) \, dz \leq \frac{s}{\gamma} \int_{\mathbb{R}} S(\ell) \, d\ell = \frac{2s}{\gamma}.$$

The velocity factor integrates to one. Thus the mass of each annular sign prototype is at most Cs^2 . Its velocities satisfy

$$|v| \leq 1 + \alpha + d_s < 2,$$

which gives the same bound for its energy.

On the set where the density is positive, decompose its logarithm into the three terms

$$\log((K/s)d_s^{-3}), \quad \log S(\gamma z/s), \quad \log J(w).$$

The first has absolute value at most $C(1 + |\log s|)$. The second is integrable against its own weight since

$$\int_{\mathbb{R}} S(\ell) |\log S(\ell)| \, d\ell < \infty :$$

on $|\ell| \geq 1$, its integrand is bounded by $C(1 + |\ell|)e^{-2|\ell|}$, and on a bounded interval it is bounded. Also $\int J |\log J| < \infty$, because J is bounded and compactly supported and $r |\log r|$ is bounded for $0 \leq r \leq 1$. Consequently the absolute entropy of an annular sign prototype is at most

$$Cs^2(1 + |\log s|).$$

The sums of these bounds over the geometric scales converge. Since the spatial supports of the prototypes are disjoint modulo periodicity, the same estimates prove convergence of the truncated mass, energy, and absolute entropy. As $-\log M$ is a constant plus $|v|^2/2$, they prove relative entropy convergence as well.

For the seed $h_N^0 = \lambda_N s_N \chi(x/s_N, v)$, a change of spatial variables gives

$$\int h_N^0 (1 + |v|^2) \leq Cs_N^4, \quad \int h_N^0 |\log h_N^0| \leq Cs_N^4 (1 + |\log s_N|).$$

The constants are uniform in $\lambda_N \in [0, L_0]$, including $\lambda_N = 0$, since $\lambda |\log \lambda|$ is bounded on bounded intervals. It remains to check overlap with the cold datum. At every point

$$0 \leq F_0^{(N)} \leq Cs_N^{-31}.$$

For $f, b \geq 0$, integration of $1 + \log(f + r)$ over $0 < r < b$ gives, with the convention at $b = 0$,

$$|(f + b) \log(f + b) - f \log f| \leq b\{2 + \log^+(f + b)\} + b|\log b|.$$

Indeed the positive part of the logarithm is bounded by $\log^+(f + b)$, and $\int_0^b \log^-(f + r) \, dr \leq \int_0^b \log^- r \, dr \leq b(1 + |\log b|)$. On the seed support, $\log^+(F_0^{(N)} + h_N^0) \leq C(1 + |\log s_N|)$. The entropy perturbation is therefore $O(s_N^4(1 + |\log s_N|))$. The mass and Maxwellian-reference terms are controlled by the seed's mass and energy estimates. This proves the last assertion. $\square$

4.2 A transported Maxwellian barrier

For a fixed scale and sign, the equation

$$u_s^\sigma(t, z) = a_s^\sigma(z - tu_s^\sigma(t, z)) \quad (4.5)$$

defines u_s^σ for all $t \geq 0, z \in \mathbb{R}$. Indeed $\tau \mapsto \tau + ta_s^\sigma(\tau)$ is strictly increasing, has derivative at least one, and maps $\mathbb{R}$ onto $\mathbb{R}$. In particular, $|u_s^\sigma| \leq 1 + \alpha$. For label $i = (j, \sigma, n)$, abbreviate

$$u_i(t, z) = u_{s_j}^\sigma(t, z - n_3), \quad \tau_i = z - n_3 - tu_i(t, z), \quad w_i = v - u_i(t, z)e_3,$$

and set

$$D_{y,i} = \text{dist}(y - \bar{n}, E_j^\sigma), \quad D_{z,i} = \text{dist}(\tau_i, [-a, a]).$$

Choose $c_{\text{sp}} = 1/8$ and a fixed sufficiently large C_m , depending only on J , and define, for $t > 0$,

$$m_i(t, x, v) = C_m \frac{K}{s_j} d_{s_j}^{-3} e^t \exp \left[-\frac{(2-t)|w_i|^2}{d_{s_j}^2} - \frac{c_{\text{sp}}(D_{y,i}^2 + D_{z,i}^2)}{d_{s_j}^2 t} \right]. \quad (4.6)$$

Its trace at $t = 0$ is defined by the limit along free characteristics. Boundary values of the spatial masks are irrelevant. Denote the free transport of the i -th initial color by

$$f_i(t, x, v) = c_i^0(x - tv, v).$$

Lemma 4.3 (Barrier identities). *The constant C_m can be chosen so that $f_i \leq m_i$ on $[0, \bar{T}]$, uniformly in the label. Moreover, almost everywhere along every free characteristic,*

$$D_t m_i \geq \left(1 + \frac{7}{8}|w_i|^2/d_{s_j}^2\right) m_i. \quad (4.7)$$

There is a constant C_ν , independent of the label, such that

$$P(m_i, m_i) = m_i \nu_{m_i}, \quad \nu_{m_i}(v) \leq C_\nu \frac{K}{s_j} (d_{s_j} + |w_i|). \quad (4.8)$$

Every free core remains in its own spatial enlargement:

$$f_i(t, x, v) > 0 \implies x \in \mathcal{O}_i(t).$$

Proof. Suppress the indices, write $d = d_s$, and put $A = (a_s^\sigma)'(\tau) \geq 0$. Implicit differentiation gives

$$u_z = \frac{A}{1 + tA} \geq 0, \quad u_t = -u u_z, \quad D_t u = (v_3 - u)u_z, \quad D_t \tau = \frac{v_3 - u}{1 + tA}. \quad (4.9)$$

Consequently

$$D_t |w|^2 = -2u_z w_3^2.$$

The distance functions are Lipschitz, hence absolutely continuous along flights, and at their almost-everywhere differentiability points

$$|D_t D_y| \leq |\bar{w}|, \quad |D_t D_z| \leq |w_3|/(1 + tA) \leq |w_3|.$$

Differentiating the logarithm of (4.6) now gives

$$\begin{aligned} D_t \log m = 1 + \frac{|w|^2}{d^2} + \frac{2(2-t)u_z w_3^2}{d^2} \\ + \frac{c_{\text{sp}}(D_y^2 + D_z^2)}{d^2 t^2} - \frac{2c_{\text{sp}}(D_y D_t D_y + D_z D_t D_z)}{d^2 t}. \end{aligned}$$

For each of the two spatial terms use $r^2 - 2rq \geq -q^2$. This yields

$$D_t \log m \geq 1 + (1 - c_{\text{sp}})|w|^2/d^2 + 2(2-t)u_z w_3^2/d^2.$$

The last term is nonnegative on the preliminary time interval, proving (4.7). In particular the large derivative of the initial axial profile causes no adverse s^{-1} term.

To compare with free transport, start a core flight at $x^0 = (y^0, z^0)$ in its initial support and write

$$v = a_s^\sigma(z^0 - n_3)e_3 + dw^0, \quad |w^0| \leq 1.$$

At $x = x^0 + tv$, the function $p \mapsto p - a_s^\sigma(z - n_3 - tp)$ has derivative at least one, its value at v_3 is dw_3^0 , and its value at u is zero. Thus

$$|v_3 - u| \leq d|w_3^0|, \quad |\bar{v}| = d|\bar{w}^0|, \quad D_y \leq td|\bar{w}^0|, \quad D_z \leq td|w_3^0|.$$

In particular $|w| \leq d$ and $(D_y^2 + D_z^2)/(d^2 t) \leq t$. The exponential in (4.6), including e^t , is bounded below on this flight by e^{-3} . Taking $C_m \geq e^3 \|J\|_\infty$ gives $f_i \leq m_i$, since $S \leq 1$. The same calculation supplies the required initial domination. Away from the closed initial spatial support the trace of m_i is zero; on it the distance exponent tends to zero along flights. The inequalities therefore integrate from time zero. No differentiation of a spatial indicator is used.

At a fixed point (t, x) , m_i is a scalar multiple of the Maxwellian with center $u_i e_3$ and exponent $(2-t)/d_s^2$. Conservation of momentum and energy in an elastic collision gives $m'_i m'_{i,*} = m_i m_{i,*}$, which proves the first identity in (4.8). Its velocity mass is at most CK/s , and its first moment about $u_i e_3$ is at most CKd_s/s , by the substitution $p = u_i e_3 + d_s \xi$. The inequality $|v - p| \leq |w_i| + |p - u_i e_3|$ proves the frequency bound.

Finally, the free-flight estimates and (4.3) give

$$D_y \leq \bar{T}d_s \leq \delta_j/2.$$

Its axial velocity has modulus less than 2, so its position lies between $n_3 - a - 2t$ and $n_3 + a + 2t$. It belongs to $\mathcal{O}_i(t)$, as asserted. $\square$

4.3 Cold control and the density Jacobian

The labels for the finite approximation are

$$\mathcal{I}_N = \{(j, \sigma, n) : 1 \leq j \leq N, \sigma = \pm 1, n \in \mathbb{Z}^3\}.$$

For $i = (j, \sigma, n)$ we also write $s_i = s_j$. The infinitely many integer-copy labels merely implement periodicity: translating x permutes them. We consider nonnegative colored mild solutions with these initial colors and any nonnegative hot component, with a common finite collision cutoff $b > 0$. The statement below applies throughout every interval on which the finite-approximation solution exists with bounded weighted sup norm. Its constants do not use that norm or any bound on the hot component.

Proposition 4.4 (Uniform cold control). *There is a constant L , depending on the fixed parameters, such that after imposing the additional condition*

$$Ld_0\bar{s}^9 \leq 1, \quad (4.10)$$

the following statements hold at every time of existence of the finite approximation in $0 \leq t \leq \bar{T}$.

(i) *For every retained label $i = (j, \sigma, n)$,*

$$0 \leq c_i \leq f_i + \varepsilon_j m_i, \quad \varepsilon_j = Ld_{s_j}/s_j. \quad (4.11)$$

(ii) *At each spatial point call the unique label with $x \in \mathcal{O}_i(t)$, if one exists and is retained, the main label. Let c_{main} be that color, or zero, and set $e = c - c_{\text{main}}$. For every $A \geq 0$,*

$$0 \leq e(t, x, v) \leq CA^A t^A e^{-4|v|^2}. \quad (4.12)$$

The upper bound remains valid after restricting the escaped sum to any subset of its labels.

(iii) *If the main label has scale s , its velocity measure is bounded by a fixed multiple of $(t + s)^{-1}$ times a sum of admissible probability sheets from Lemma 3.4. Its free-core contribution has the exact parameterization (4.16) below. The remainder has the smaller velocity-mass coefficient Cd_s/s^2 . In particular, for every fixed $k \geq 0$,*

$$\int_{\mathbb{R}^3} \langle v \rangle^k c_{\text{main}} \, dv \leq \frac{C_k}{t + s}, \quad \int_{|v| > R} \langle v \rangle^k c_{\text{main}} \, dv \leq \frac{C_k e^{-c_k R^2}}{t + s} \quad (R \geq 3). \quad (4.13)$$

(iv) *For the mixed-cold source in the hot equation,*

$$\sum_{\substack{i, j \in \mathcal{I}_N \\ i \neq j}} P^b(c_i, c_j)(t, x, v) \leq CA^A t^{A-1} H(v) \quad (t > 0) \quad (4.14)$$

for every $A \geq 0$, with constants uniform in N and b .

Proof. Nonlinear comparison. For one color put $U_i = f_i + \varepsilon_j m_i$. Lemma 4.3 gives $U_i \leq (1 + \varepsilon_j)m_i$. By (4.10), $\varepsilon_j \leq 1$; therefore

$$P^b(U_i, U_i) \leq P(U_i, U_i) \leq 4C_\nu \frac{K}{s_j} (d_{s_j} + |w_i|) m_i.$$

On the other hand,

$$D_t U_i \geq L \frac{d_{s_j}}{s_j} (1 + \frac{7}{8} |w_i|^2 / d_{s_j}^2) m_i.$$

The ratio $(1 + r)/(1 + \frac{7}{8} r^2)$ is bounded on $r \geq 0$. Choose $L \geq 1$ at least $4C_\nu K$ times that bound. Then $D_t U_i \geq P^b(U_i, U_i)$. This choice is made before decreasing d_0 , so (4.10) is compatible with all earlier constraints.

The cold equation implies $D_t c_i \leq P^b(c_i, c_i)$, because its loss is nonnegative. To justify comparison explicitly, integrate both inequalities along a free path, using $c_i(0) \leq U_i(0)$, and put $z_i = (c_i - U_i)_+$. Positivity and bilinearity give

$$z_i(t) \leq \int_0^t [2P^b(U_i, z_i) + P^b(z_i, z_i)](\tau, x - (t - \tau)v, v) \, d\tau.$$

On any interval of bounded weighted norm, the gain bound Lemma 3.7 implies

$$\|z_i(t)/H\|_\infty \leq C_I \int_0^t \|z_i(\tau)/H\|_\infty d\tau.$$

Indeed m_i/H is bounded for a fixed scale, uniformly on the preliminary interval, and $z_i \leq c_i + U_i$ supplies the other bounded factor. Iteration of this integral inequality, or its scalar integrating factor, gives $z_i = 0$. This proves (4.11) without a hot bound. The argument is unchanged for each periodic-copy label.

Summation of escaped tails. For a nonmain label, (4.2) and $|u_i| \leq 1 + \alpha < 2$ imply either $D_{y,i} \geq \delta_j$ or $D_{z,i} \geq b_{\text{sp}}$. Since $\delta_j \leq b_{\text{sp}}$, in either case

$$D_{y,i}^2 + D_{z,i}^2 \geq (c_\eta/4)^2 s_j^4.$$

If $a_v = (2-t)/d_s^2$, then $a_v \geq 8$, and completing the velocity square gives

$$\sup_v \exp\{4|v|^2 - a_v|v - ue_3|^2\} = \exp\left(\frac{4a_v|u|^2}{a_v - 4}\right) \leq e^{8(1+\alpha)^2}.$$

Thus $c_i \leq 2m_i$ bounds its escaped contribution by

$$Cs_j^{-31} e^{-4|v|^2} \exp\left[-\frac{c_{\text{sp}}(D_{y,i}^2 + D_{z,i}^2)}{d_s^2 t}\right]. \quad (4.15)$$

The factor d_0^{-3} has been included in the fixed constant.

For completeness the integer-copy sum is uniform in the scale. Since $E_j^\sigma \subset [-s_0, s_0]^2$,

$$D_{y,i}^2 \geq \sum_{k=1}^2 (|y_k - n_k| - s_0)_+^2, \quad D_{z,i} \geq (|z - n_3| - a - 2\bar{T})_+.$$

For $\beta > 0, R \geq 0, \epsilon > 0$, comparison of the two monotone Gaussian tails with their integrals gives

$$\sup_{\xi \in \mathbb{R}} \sum_{n \in \mathbb{Z}} e^{-\beta(|\xi - n| - R)_+^2 / \epsilon^2} \leq 2R + 5 + \epsilon \sqrt{\pi/\beta}.$$

There are at most $2R + 2$ integers in the central interval; for each of the remaining tails, keep its first term and bound the rest by the corresponding Gaussian integral, which proves the displayed bound. Applying this estimate to the three coordinates, with $\epsilon = d_s \sqrt{t}$, bounds the lattice sum of one half of the spatial exponent uniformly for $t \leq \bar{T}$. Use the other half to retain the escaped distance. For some fixed $\beta_0 > 0$ this gives

$$\sum_{n: i \text{ nonmain}} e^{-c_{\text{sp}}(D_{y,i}^2 + D_{z,i}^2)/(d_s^2 t)} \leq C \exp[-\beta_0/(s^{16}t)].$$

Here β_0 includes d_0^{-2} , and the two signs only contribute a factor two.

For arbitrary $A \geq 0$ and $B > 0$, choose $p \geq A$ with $16p - 31 \geq B$. From $e^{-x} \leq (p/e)^p x^{-p}$, and $t \leq 1$,

$$s^{-31} e^{-\beta_0/(s^{16}t)} \leq C_{A,B} t^A s^B.$$

The geometric sum of s_j^B is finite. More precisely, restricting to $s_j \leq \rho$ gives a bound $C_{A,B} t^A \rho^B$. This proves (4.12); it also shows that any fixed extra power of s_j^{-1} can be absorbed by increasing

p . All these estimates were obtained by summing upper bounds over every scale and copy, so they are uniform in N .

The exact free-core velocity measure. Fix t, x and one label, abbreviating $z_c = z - n_3$, $y_c = y - \bar{n}$, $s = s_j$, $d = d_s$. For $w \in \mathbb{R}^3$ define $\Phi(w_3)$ by

$$\Phi(w_3) = a_s^\sigma(z_c - t\Phi(w_3)) + dw_3, \quad \bar{p} = d\bar{w}, \quad p_3 = \Phi(w_3).$$

The same strictly increasing map used in (4.5) gives a unique Φ , and

$$\Phi'(w_3) = \frac{d}{1 + t\alpha\gamma S(\gamma(z_c - t\Phi(w_3))/s)/s}.$$

The full velocity change of variables therefore gives exactly

$$\begin{aligned} f_i(t, x, p) \, dp &= \frac{KS(\gamma(z_c - tp_3)/s)}{s + t\alpha\gamma S(\gamma(z_c - tp_3)/s)} J(w) \\ &\quad \times \mathbf{1}_{E_j^\sigma}(y_c - td\bar{w}) \mathbf{1}_{(-a, a)}(z_c - tp_3) \, dw. \end{aligned} \quad (4.16)$$

For $k_0 = \alpha\gamma > 0$ and $0 < S \leq 1$,

$$\frac{KS}{s + tk_0S} \leq \frac{K}{s + tk_0} \leq \frac{K \max(1, k_0^{-1})}{s + t}. \quad (4.17)$$

Thus this estimate is uniform even when t/s is large or the value of S is small.

We spell out the product-sheet upper bound, since the axial law can be very compressed. Let

$$g_2(\bar{w}) = \pi^{-1} e^{-|\bar{w}|^2}, \quad g_1(w_3) = \pi^{-1/2} e^{-w_3^2}.$$

There is a fixed C_J such that $J(w) \leq C_J g_2(\bar{w}) g_1(w_3)$. The axial map Φ depends only on w_3 , and

$$|\Phi(w_3) - u_s^\sigma(t, z_c)| \leq d|w_3|$$

by the derivative-at-least-one argument. The pushforward of $g_2(\bar{w}) \, d\bar{w}$ by $\bar{w} \mapsto d\bar{w}$ is a centered transverse Gaussian. Independently, let ρ be the pushforward of $g_1(w_3) \, dw_3$ by Φ . It is a probability law and, uniformly in s, t, x ,

$$\int_{\mathbb{R}} e^{8p_3^2} \rho(\, dp_3) \leq e^{16(1+\alpha)^2} \int_{\mathbb{R}} e^{16d^2 w_3^2} g_1(w_3) \, dw_3 \leq C.$$

Equations (4.16) and (4.17), after discarding the masks, bound the core measure by $C/(t+s)$ times this product law. No density bound on ρ is asserted or needed.

For the remainder, use $p = u_i e_3 + d_s w$ in (4.6). Its measure is bounded by

$$\varepsilon_j m_i \, dp \leq C \frac{d_s}{s^2} e^{-(2-t)|w|^2} \, dw,$$

which is again a product-sheet majorant, with the stated smaller coefficient. Since

$$(d_s/s^2)(t+s) = d_0 s^8(t+s) \leq d_0 \bar{s}^8(\bar{T} + \bar{s}),$$

this coefficient is also at most $C/(t+s)$. The width restrictions in (4.3) allow these Gaussian laws and their fixed, slightly wider majorants in Lemma 3.4. Lemma 3.6 applies with the common

axial bound $A = 1 + \alpha$, the fixed Gaussian majorant constant, and the displayed physical widths. The factors $C/(t + s)$ and Cd_s/s^2 remain outside the probability-sheet normalization. A parameter restriction $|w| > R$ therefore contributes at most Ce^{-cR^2} times its respective amplitude to the normalized gain. For the remainder, $2 - t \geq 2 - \bar{T} > 1$, so its parameter Gaussian is bounded by $e^{-|w|^2}$. The normalized kernel-tail conclusions of Lemma 3.5 also apply, with these amplitudes.

The core velocities are supported in $|p| < 2$. The remainder has uniformly small Gaussian width and a center in $|p_3| \leq 1 + \alpha$. Substitution in its Gaussian integral gives all its polynomial moments and the uniform Gaussian moment tails. Combined with (4.17), these give (4.13).

The mixed source. At a fixed point, separate the main color from e . Positivity and the ordered nature of the cross-color sum give

$$\sum_{i \neq j} P^b(c_i, c_j) \leq 2P(c_{\text{main}}, e) + P(e, e).$$

The function $e^{-4|v|^2}/H(v)$ is essentially bounded, so (4.12) bounds e by $C_A t^A H$. Lemma 3.4 and the product measures just proved bound $P(c_{\text{main}}, H)$ by $C(t + s)^{-1}H$; when there is no main color this term is zero. Lemma 3.7 bounds $P(H, H)$ by CH . Since $t + s \geq t$ and $t \leq 1$,

$$2P(c_{\text{main}}, e) + P(e, e) \leq C_A \{t^A/(t + s) + t^{2A}\}H \leq C'_A t^{A-1}H.$$

This proves (4.14). $\square$

It is useful to record the spatial form of the moment bound. For a fixed $k \geq 0$, Proposition 4.4 implies

$$\int_{\mathbb{R}^3} \langle v \rangle^k c(t, x, v) \, dv \leq C_k \sum_{i \in \mathcal{I}_N} \frac{\mathbf{1}_{\mathcal{O}_i(t)}(x)}{t + s_i} + C_{A,k} t^A. \quad (4.18)$$

At most one term of the displayed sum is present at any point. In one periodic cell, the spatial volume of the enlargement belonging to a fixed annular sign, with its periodic copies included, is at most Cs_i^2 . The same bound with a factor tending to zero as the velocity cutoff increases holds for the integrated velocity tails.

4.4 Quantitative survival on a free core

The upper estimates did not use a hot bound. The following consequence, used only for the dilation, adds the cap assumption.

Corollary 4.5 (Survival of a cold core under the cap). *Suppose a finite approximation satisfies*

$$0 \leq h(t, x, v) \leq (t + s_N)H(v)$$

on an initial interval $[0, T] \subset [0, \bar{T}]$. All conclusions below concern times $t \in [0, T]$. For every $M \geq 0$ there are constants C, C_M , independent of T, N , the cutoff, and the seed, such that

$$\begin{aligned} e^{-E_{s,N}(t)} f_i(t, x, v) &\leq c_i(t, x, v) \leq f_i(t, x, v) + L(d_s/s) m_i(t, x, v), \\ E_{s,N}(t) &= C(d_s t/s + t^2 + s_N t) + C_M t^{M+1} \end{aligned} \quad (4.19)$$

for every label of scale s . Consequently, if $r \rightarrow 0$, $s_N \leq r$, $s/r \in [\epsilon, R]$ for fixed $0 < \epsilon < R < \infty$, and $0 \leq t \leq \min(T, Ar)$, then

$$E_{s,N}(t) \leq C_{\epsilon,R,A}(d_s + r^2) + C_{M,A} r^{M+1} \rightarrow 0$$

uniformly in the label and in points of its free core. The transverse free displacement divided by the stripe period satisfies

$$\frac{td_s}{s^2} \leq C_{\epsilon, R, A} r^9 \longrightarrow 0. \quad (4.20)$$

On these scales the difference between cold and free-core velocity measures also satisfies, uniformly in position and time,

$$r \int_{\mathbb{R}^3} |c_i - f_i|(t, x, v) \, dv \longrightarrow 0. \quad (4.21)$$

Proof. The loss-integrating-factor formula for the i -th mild equation and nonnegativity of its gain give

$$c_i(t, x, v) \geq f_i(t, x, v) \exp \left[- \int_0^t \nu_F^b(\tau, x - (t - \tau)v, v) \, d\tau \right].$$

Only points with $f_i(t, x, v) > 0$ need be considered. Their entire backward flight, equivalently the corresponding forward flight from the initial support, is in $\mathcal{O}_i(\tau)$ by Lemma 4.3. Also $|v - u_i(\tau, z(\tau))e_3| \leq d_s$ throughout it.

At such a position the free part of the same color has velocities within d_s of this center, by (4.16), and mass at most $C/(\tau + s)$. Its contribution to the frequency at v is at most $Cd_s/(\tau + s) \leq Cd_s/s$. The remainder has mass Cd_s/s^2 and first moment about the center at most Cd_s^2/s^2 . Since $d_s/s \leq 1/L$, its contribution is at most Cd_s/s as well. The same-color frequency is therefore bounded by Cd_s/s .

Every other color is escaped along this flight. Its total frequency is at most $C_M \tau^M$, by (4.12) and $|v| < 2$. The cap applies at every preceding time $\tau \in [0, t]$, so

$$\nu_h(v) \leq 2\pi(\tau + s_N) \int_{\mathbb{R}^3} (|v| + |p|) H(p) \, dp \leq C(\tau + s_N)$$

at these bounded velocities. Since $\nu_F^b \leq \nu_F$, integration proves the optical-depth bound in (4.19). Its upper bound is (4.11).

If $s/r \in [\epsilon, R]$, $t \leq Ar$, and $s_N \leq r$, the first term in the optical depth is bounded by $C_{A, \epsilon} d_s$, and the two hot terms are $O_A(r^2)$. All stated convergence is uniform. Moreover

$$td_s/s^2 = d_0 t s^8 \leq d_0 A R^8 r^9,$$

which proves (4.20).

For the last claim the sandwich gives

$$|c_i - f_i| \leq (1 - e^{-E_{s, N}(t)}) f_i + L(d_s/s) m_i.$$

Integrating in velocity and using the core and remainder density bounds yields

$$r \int |c_i - f_i| \, dv \leq C E_{s, N}(t) \frac{r}{t + s} + C \frac{r d_s}{s^2}.$$

Here $r/(t + s) \leq 1/\epsilon$, and $r d_s/s^2 = d_0 r s^8 \leq C_R r^9$. Both terms tend to zero uniformly, proving (4.21). $\square$

5 Finite approximations and cap selection

Retain the first N annuli of the datum in (4.4), and use a common collision cutoff $b_N \rightarrow \infty$ in (2.4). All assertions in this section first concern fixed N, b_N . Constants in the cold estimates and in the caps will be uniform in N .

Choose once and for all a nonnegative smooth function $\chi \in C_c^\infty(\mathbb{R}_x^3 \times \mathbb{R}_v^3)$, not identically zero. After reducing s_0 , its periodic copies at spatial scale s_N are disjoint initially. The hot initial density is

$$h(0, x, v) = \lambda s_N \chi(x/s_N, v),$$

periodized in x . The cold initial densities are independent of λ . In particular the norm $\|\chi(x/s_N, v)\|_H$ is a fixed positive number independent of N .

Proposition 5.1. *There is a fixed $T_0 > 0$ such that the following statements hold.*

- (a) *With $\lambda = 0$, the finite approximations exist on $[0, T_0]$ and satisfy $0 \leq h \leq t^2 H$.*
- (b) *There is a fixed finite L_0 such that, for every $0 < T \leq T_0$ and every N , some $\lambda \in [0, L_0]$ gives an approximation on $[0, T]$ with*

$$0 \leq h(t, x, v) \leq (t + s_N)H(v), \quad \max_{0 \leq t \leq T} \frac{\|h(t)\|_H}{t + s_N} = 1. \quad (5.1)$$

In both cases all cold components obey Proposition 4.4.

Proof. We first give the fixed-approximation local theory used in the selection. Lemma 3.7 bounds the gain bilinearly in $\|\cdot\|_H$. For the cutoff loss,

$$\|\nu_f^{b_N}\|_\infty \leq 2\pi b_N \operatorname{ess\,sup}_x \int_{\mathbb{R}^3} |f(x, w)| \, dw \leq 2\pi b_N \|f\|_H \int_{\mathbb{R}^3} H(w) \, dw,$$

so gain and loss are locally Lipschitz on bounded balls of the weighted norm. Free transport is an isometry in that norm.

Picard iteration for the integral equation along characteristics gives a unique local solution in the space of bounded measurable functions with finite weighted norm. An integrating factor for the loss, followed by positive iteration of the gain, gives nonnegativity. One can equivalently perform this iteration for the labeled system, using the norm of the sum of the absolute values of all components, and for differences the sum of the absolute differences. The quadratic estimates close in that norm. The Gaussian spatial tails in Proposition 4.4 make the sums over periodic copies finite in the relevant norm. Periodic covariance is preserved by uniqueness.

The solution continues as long as its total weighted norm remains bounded. It depends continuously on its initial datum, uniformly on a fixed interval on which that norm is bounded. These assertions follow from the same integral contraction and its difference estimate on short subintervals. They do not require norm continuity of spatial translations: one works with the integral equation along the free paths. In particular,

$$\|h(t) - h(s, x - (t - s)v, v)\|_H \leq C_{N, b_N} |t - s|$$

on a bounded solution interval. Since transport is an isometry, $t \mapsto \|h(t)\|_H$, and hence $t \mapsto \|h(t)\|_H / (t + s_N)$, are continuous.

Let $z(t) = \|h(t)\|_H$. Dropping the loss in the hot equation, the cold and kernel estimates give its scalar integral comparison with

$$z'(t) \leq \left(\frac{C_1}{t + s_N} + C_1 \right) z(t) + C_1 z(t)^2 + C_A t^{A-1}. \quad (5.2)$$

Here A can be any sufficiently large fixed number, and its source constant is uniform in N . This is an upper integral inequality; no differentiability of the essential supremum is needed.

Fix $A > C_1 + 2$. Choose $D > 0$ so that $(A - C_1)D > C_A$. Since $t/(t + s_N) \leq 1$, the function Dt^A is a supersolution of (5.2) after shortening a fixed preliminary time interval to absorb the bounded linear and quadratic terms. Shorten it again to ensure $Dt^A \leq t^2$. Comparison from zero proves (a). The cold bound is independent of h , so it also ensures continuation for each fixed N, b_N up to this time.

For (b), shorten T_0 to be less than one, so that the zero-seed solution satisfies a strict cap:

$$\frac{\|h(t)\|_H}{t + s_N} \leq \frac{t^2}{t + s_N} \leq T_0 < 1.$$

Choose L_0 so large that $L_0 \|\chi(x/s_N, v)\|_H > 1$. The cap then fails at time zero for $\lambda = L_0$.

For fixed N, b_N, T , let $\mathcal{O}$ be the set of seed parameters in $[0, L_0]$ whose solution exists up to T and obeys a strict cap there. Continuous dependence and continuation make $\mathcal{O}$ relatively open. Its component containing zero has a right endpoint λ_* strictly below L_0 : indeed a neighborhood of L_0 already fails the initial cap. Approach λ_* from within that component. The cap and the fixed cold bound give a common finite weighted norm on $[0, T]$, with constants permitted to depend on N, b_N . The local difference estimates on finitely many short subintervals give a limit solution at λ_* , with cap at most one. It continues to T . If its maximum cap ratio were less than one, continuous dependence would extend the component past λ_* , a contradiction. This proves (5.1). $\square$

Remark 5.2. The coefficient C_1 may be large, in particular when $\alpha\gamma$ is small. It is fixed before A, D, T_0 are chosen. The arbitrary-power tail estimates are available on a preliminary short interval; their constants can be fixed there before the final shortening. Thus the choice does not depend on N , and imposes no restriction on the later choice of shrinking horizons.

6 Fixed-horizon limits and admissibility

In this section the physical horizon $T \in (0, T_0]$ is fixed. The approximation index tends to infinity without any rescaling of time or space. This distinction matters: the cold densities in this section have uniformly integrable limits, whereas their rescalings in the next section will converge to measures in velocity.

Proposition 6.1. *Fix $T \in (0, T_0]$. Either the zero-seed approximations of Proposition 5.1, or any sequence of its cap shots on $[0, T]$, has a subsequence whose total densities $F_N = c_N + h_N$ converge weakly in $L^1([0, T] \times \Omega)$ to a nonnegative density F . The subsequence can be chosen so that $F_N(t) \rightharpoonup F(t)$ in $L^1(\Omega)$ at every $t \in [0, T]$, with uniformly small polynomially weighted velocity tails. The limit has the common initial datum (4.4), and*

$$Q^\pm(F, F) \in L^1([0, T] \times \Omega), \quad F \in C([0, T]; L^1(\Omega)).$$

It satisfies every condition of Definition 2.1 on this interval, including the inequalities at every time. Each such limit has an admissible global continuation. The dormant upper bound, consisting of the cold majorants of Proposition 4.4 and $t^2 H$, passes to a zero-seed limit.

We give the compactness and entropy arguments in detail.

6.1 Conservation and entropy before passage to the limit

At a fixed approximation the bounds already proved imply $F_N \leq A_N H$ on the whole solution interval, with a finite constant allowed to depend on N, b_N, T . In particular there are all polynomial velocity moments, and there is an L^p majorant for some $1 < p < 2/q$. The equation holds absolutely continuously on almost every free path. Its cutoff gain and loss are integrable, including after multiplication by any fixed polynomial in velocity.

The collision map preserves volume, kinetic energy, and $|v - v_*|$, so it also preserves the cutoff. For an integrable velocity test ψ the collision symmetry gives

$$\int_{\mathbb{R}^3} \psi(v) Q^{b_N}(F_N, F_N)(v) dv = \frac{1}{2} \int_{\mathbb{R}^3 \times \mathbb{R}^3 \times S^2} |a| \mathbf{1}_{\{|v-v_*| < b_N\}} F_N F_{N,*} (\psi' + \psi'_* - \psi - \psi_*) dv dv_* d\omega.$$

The right side vanishes for $\psi = 1, v_j, |v|^2$. Integrating the path equation and using periodicity therefore proves mass, momentum, and energy conservation at every time of the finite approximation. The same calculation with a spatially dependent mass test gives its local continuity equation.

To justify entropy at zeros, put $E(v) = e^{-2|v|^2}$, take $0 < \delta \leq 1$, and set $F_{N,\delta} = F_N + \delta E$. Define

$$\mathcal{A}_\delta(f, v) = M(v) + \int_0^f \log \frac{u + \delta E(v)}{M(v)} du.$$

Its derivative with respect to f is $\log(F_{N,\delta}/M)$. At the fixed approximation,

$$\left| \log \frac{F_{N,\delta}}{M} \right| \leq C_N (1 + |\log \delta| + |v|^2 + \log^+ |\bar{v}|^{-1}). \quad (6.1)$$

Indeed $F_{N,\delta} \geq \delta e^{-2|v|^2}$ gives the lower logarithmic bound, and $F_N \leq A_N H$ gives the upper bound. Every factor on the right of (6.1) is integrable against H , apart from the harmless constant $|\log \delta|$: in the transverse plane this uses $\int_0^1 r^{1-q} |\log r| dr < \infty$.

Since $E \leq CH$, the fixed-cutoff bilinear bounds imply

$$|Q^{b_N}(F_{N,\delta}, F_{N,\delta}) - Q^{b_N}(F_N, F_N)| \leq C_{N,b_N} \delta H.$$

Consequently replacing the latter operator by the former in its integral against $\log(F_{N,\delta}/M)$ costs at most $C_{N,b_N,T} \delta (1 + |\log \delta|)$. The path chain rule, followed by spatial integration, now gives

$$\int_{\Omega} \mathcal{A}_\delta(F_N(t), v) + \int_0^t \mathcal{D}_{b_N}(F_{N,\delta}(s)) ds \leq \int_{\Omega} \mathcal{A}_\delta(F_N(0), v) + C_{N,b_N,T} \delta (1 + |\log \delta|). \quad (6.2)$$

Here $\mathcal{D}_b$ denotes the dissipation with the invariant cutoff inserted. The collision identity used in this step is

$$\int_{\Omega} Q^{b_N}(F_{N,\delta}, F_{N,\delta}) \log(F_{N,\delta}/M) = -\mathcal{D}_{b_N}(F_{N,\delta}).$$

It follows by averaging over interchange of the two particles and over the elastic substitution. The $\log M$ term cancels because it is a linear combination of mass and energy. All integrals for this calculation are finite at $\delta > 0$ by (6.1).

As $\delta \downarrow 0$, the primitive converges to $f \log(f/M) - f + M$. Its integrals converge as well. For example its lower bound is that limiting primitive, while its upper bound follows by replacing δ by one in the increasing integrand. The negative part of $f \log f$ is integrable under a finite second moment, and the upper bound is integrable by (6.1) without the $|\log \delta|$ term. Fatou's lemma applied to the nonnegative dissipation integrand in (6.2) therefore yields, for every $t \in [0, T]$,

$$\mathcal{H}_M(F_N(t)) + \int_0^t \mathcal{D}_{b_N}(F_N(s)) ds \leq \mathcal{H}_M(F_N(0)). \quad (6.3)$$

Lemma 4.2 gives convergence of the initial mass, energy, and relative entropy to those of (4.4). Its estimates include the hot seed, whose mass and energy are $O(s_N^4)$ and whose absolute entropy is $O(s_N^4(1 + |\log s_N|))$, as well as the seed's overlap with the cold density in the relative entropy. The convergence is uniform for the allowed seed parameters $0 \leq \lambda \leq L_0$.

6.2 Uniform integrability of densities and collision products

For $s = s_j$, let A_s be the annulus $\{y \in \mathbb{R}^2 : s < |y| < e^\eta s\}$, viewed as a subset of $\mathbb{T}^2$. By Lemma 4.1, these annuli are pairwise disjoint and contain the enlarged stripes of both signs at their respective scales. Their areas satisfy $|A_s| \leq Cs^2$. Proposition 4.4, the cap, and the dormant bound imply, for every fixed $k \geq 0$,

$$R_{k,N}(t, x) := \int_{\mathbb{R}^3} \langle v \rangle^k F_N(t, x, v) dv \leq C_k \left(1 + \sum_{j \leq N} \frac{\mathbf{1}_{A_{s_j}}(y)}{t + s_j} \right). \quad (6.4)$$

The bounded term includes the hot density and escaped tails. The cold remainder has density coefficient $O(d_s/s^2) = O(s^8)$, and is also covered by this estimate. All constants are independent of N .

The square of the envelope is integrable uniformly in N , including at the initial surface. In fact the contribution of one annulus is

$$\int_0^T \frac{Cs^2}{(t+s)^2} dt = Cs^2 \left(\frac{1}{s} - \frac{1}{T+s} \right) \leq Cs. \quad (6.5)$$

Since the scales form a geometric sequence, $\sum_{s_j < \varepsilon} s_j \leq C_\eta \varepsilon$. If $A_{<\varepsilon} = \bigcup_{s_j < \varepsilon} A_{s_j}$, disjointness of the annuli thus gives

$$\sup_N \int_0^T \int_{A_{<\varepsilon} \times \mathbb{T}} R_{k,N}(t, x)^2 dx dt \longrightarrow 0 \quad (\varepsilon \downarrow 0). \quad (6.6)$$

The contribution of the bounded term is at most $CT|A_{<\varepsilon}|$.

Outside $A_{<\varepsilon}$, the main color has one of finitely many scales $s_j \geq \varepsilon$. Those velocity widths are positive and fixed. Their Gaussian bounds, with axial centers in a fixed bounded interval, give a function $\Psi_\varepsilon(v)$, independent of N, t, x , such that

$$F_N(t, x, v) \leq \Psi_\varepsilon(v) \quad \text{if } y \notin A_{<\varepsilon}, \quad \int \langle v \rangle^k \Psi_\varepsilon(v) dv < \infty \quad (k \geq 0). \quad (6.7)$$

One may take a finite sum of broadened Gaussians about the bounded axial interval, together with a multiple of H for the hot component and escaped tails. There is no need for this majorant to be bounded on the velocity axis.

Equations (6.6)–(6.7) prove uniform integrability of

$$\langle v \rangle^k \langle w \rangle^k F_N(t, x, v) F_N(t, x, w)$$

in the full spacetime and two-velocity variables. Indeed its integral over the removed annuli is at most the left side of (6.6); elsewhere it has the fixed integrable majorant $\langle v \rangle^k \Psi_\varepsilon(v) \langle w \rangle^k \Psi_\varepsilon(w)$. This also proves uniformly small velocity tails. Taking first powers instead proves uniform integrability of the one-particle densities in spacetime. At individual times the estimate is

$$\sup_{N, 0 \leq t \leq T} \int_{A_{<\varepsilon} \times \mathbb{T}} R_{k,N}(t, x) dx \leq C_k \sum_{s_j < C\varepsilon} \frac{s_j^2}{s_j} + C_k |A_{<\varepsilon}| \rightarrow 0. \quad (6.8)$$

Together with (6.7), this proves weighted one-particle uniform integrability and velocity tightness uniformly at all individual times.

The gains and losses enjoy the same spacetime conclusion. To see this directly, collision symmetry and energy conservation imply

$$\int \langle v \rangle^k Q_N^\pm(t, x, v) dv \leq C_k R_{k+1,N}(t, x)^2, \quad Q_N^\pm = Q^{\pm, b_N}(F_N, F_N).$$

Their weighted integrals on the small annuli therefore tend uniformly to zero. On the complement, positivity and (6.7) give the fixed bounds

$$Q_N^+(t, x, v) \leq P(\Psi_\varepsilon, \Psi_\varepsilon)(v), \quad Q_N^-(t, x, v) \leq \Psi_\varepsilon(v) \nu_{\Psi_\varepsilon}(v).$$

Both right sides are integrable with every fixed polynomial velocity weight: integrate first in the output velocity and use the preceding collision-moment inequality for the fixed function Ψ_ε . Thus $\{Q_N^\pm\}$ is uniformly integrable in spacetime, with uniformly small weighted velocity tails. In particular its integrals over time intervals of length tending to zero vanish uniformly in N . This last assertion follows by first discarding the small annuli and then integrating the fixed majorants over the time interval.

6.3 An elementary averaging argument

We use a uniformly integrable version of velocity averaging [8, Proposition 3]. The following proof gives the precise local statement needed here.

Lemma 6.2. *Suppose that f_n and g_n are locally uniformly integrable in L^1 on an open time-space-velocity region, and $(\partial_t + v \cdot \nabla_x) f_n = g_n$ there in distributions. Then for every smooth compactly supported velocity test ψ , the averages $\int \psi(v) f_n(t, x, v) dv$ are relatively compact in local $L^1_{t,x}$ on regions where the hypotheses hold for velocities in a neighborhood of $\text{supp } \psi$. The assertion applies both on a spatial torus and locally in $\mathbb{R}^3$.*

Proof. Choose smooth interior cutoffs in time, space, and velocity; the spatial cutoff is unnecessary on a torus. Write the localized sequence as q_n , extend it by zero outside the localization interval, and put $r_n = (1 + \partial_t + v \cdot \nabla_x) q_n$. The equation shows that r_n is uniformly integrable in L^1 , with support in a fixed bounded set of time and velocity and, in the Euclidean case, space.

The derivative of a spatial cutoff is harmless because velocity has also been localized. The exact identity

$$q_n(t, x, v) = \int_0^\infty e^{-s} r_n(t - s, x - sv, v) ds$$

follows by integrating the localized equation from a time preceding its support. Denote this damped inverse by R . It is an L^1 contraction, since free transport preserves Lebesgue measure.

Truncate r_n at height L , retaining its sign, to obtain $r_n^{(L)}$. Uniform integrability gives

$$\sup_n \|r_n - r_n^{(L)}\|_1 \longrightarrow 0, \quad \sup_n \|r_n^{(L)}\|_2 < \infty \quad \text{for fixed } L.$$

The averages of $R(r_n - r_n^{(L)})$ are therefore uniformly small in L^1 . It remains to handle a bounded L^2 family $r_n^{(L)}$.

Take Fourier transform in (t, x) , using Fourier series in the periodic coordinates. For $u_n = Rr_n^{(L)}$, Cauchy–Schwarz gives

$$\left| \int \psi(v) \widehat{u}_n(\tau, k, v) dv \right|^2 \leq I_\psi(\tau, k) \int |\widehat{r}_n^{(L)}(\tau, k, v)|^2 dv, \quad I_\psi(\tau, k) = \int \frac{|\psi(v)|^2}{1 + (\tau + k \cdot v)^2} dv.$$

If $\text{supp } \psi \subset \{|v| \leq R_0\}$, integration in the direction of k gives $I_\psi \leq C_\psi/|k|$ when $k \neq 0$. If $|\tau| > 2R_0|k|$, the denominator instead gives $I_\psi \leq C_\psi/(1 + |\tau|^2)$. It follows that $I_\psi(\tau, k) \rightarrow 0$ uniformly as $|\tau| + |k| \rightarrow \infty$. The averages consequently have uniformly small high-frequency L^2 tails. Convolution with a fixed smooth approximate identity approximates them uniformly in L^2 . On each compact time-space set the convolutions are bounded and equicontinuous, by Cauchy–Schwarz applied to translates of the convolution kernel and their derivatives. They are therefore relatively compact there. This proves local L^2 , hence local L^1 , compactness for fixed L . The uniform L^1 error as $L \rightarrow \infty$ proves the assertion. $\square$

6.4 Identification of the nonlinear terms

The uniform integrability just proved supplies a subsequence with $F_N \rightharpoonup F$ in spacetime L^1 ; velocity tightness allows the whole velocity space. Since $D_t F_N = Q_N^+ - Q_N^-$, Lemma 6.2 applies. After a diagonal extraction for a countable dense family of compact velocity tests, all such velocity averages converge strongly in local $L^1_{t,x}$ to the corresponding averages of F . Extension from that family to every smooth compact velocity test uses the uniform mass bound.

Set $W_N(t, x, v, w) = F_N(t, x, v)F_N(t, x, w)$. The full pair-product uniform integrability supplies a weak L^1 limit W , after another extraction. For smooth compact velocity tests ψ, χ ,

$$\int \psi(v) \chi(w) W_N dv dw = \left(\int \psi F_N dv \right) \left(\int \chi F_N dw \right).$$

Both factors converge in local L^1 , so their product converges in measure. The products are uniformly integrable in (t, x) , by (6.5) and the fixed majorants after removal of small annuli. Their convergence is consequently in L^1 : to verify this last implication, discard a set of arbitrarily small measure on which convergence is not uniform, and use uniform integrability on the discarded set. We obtain

$$\int \psi(v) \chi(w) W dv dw = \left(\int \psi F dv \right) \left(\int \chi F dw \right).$$

Finite sums of separated velocity tests are uniformly dense among continuous functions on compact two-velocity sets. Testing also in (t, x) identifies

$$W_N \rightharpoonup F(t, x, v)F(t, x, w) \quad \text{in } L^1([0, T] \times \mathbb{T}^3 \times \mathbb{R}^3 \times \mathbb{R}^3). \quad (6.9)$$

The same conclusion holds with any fixed polynomial weight for which it is needed. Interior time compactness suffices: the uniform integrability already proved removes intervals near zero and T .

Adjoining the finite angular measure preserves weak convergence. The elastic map on (v, w, ω) is measure preserving and involutive, so (6.9) also identifies the primed pair product. On a bounded two-velocity set the factor $|a|$ is bounded, and the cutoff is eventually absent. The weighted tail estimates allow these sets to exhaust all velocities. Since $Q_N^\pm$ themselves are uniformly integrable, testing their weak limits and using these identifications proves

$$Q_N^\pm \rightharpoonup Q^\pm(F, F) \quad \text{in } L^1([0, T] \times \Omega). \quad (6.10)$$

Passing the equation, including its initial boundary term, is now legitimate. The initial densities converge in L^1 , and the limit satisfies the unrenormalized distributional equation with datum F_0 .

6.5 Every-time traces, entropy, and the chain rule

We first specify representatives at every time. For a smooth periodic test $\varphi(x, v)$ compactly supported in velocity, the finite path equations give

$$|\langle F_N(t) - F_N(s), \varphi \rangle| \leq \int_s^t \int_\Omega (F_N |v \cdot \nabla_x \varphi| + (Q_N^+ + Q_N^-) |\varphi|) \, dx \, dv \, du.$$

The first term is bounded by $C_\varphi |t - s|$; the second has a uniform modulus tending to zero with $|t - s|$, by the time-integral conclusion above. For a countable uniformly dense collection of such tests, extract uniform convergence of their pairings on $[0, T]$. At each individual time (6.8) and (6.7) make $\{F_N(t)\}$ relatively weakly compact in L^1 . Every cluster point has the prescribed smooth pairings, hence is the same density. Thus this single subsequence converges weakly in L^1 at every time. It defines a weakly continuous representative against smooth tests, agreeing with the spacetime limit. The weighted fixed-time tightness also passes total mass, momentum, and energy from the finite approximations. In particular energy is conserved on this early interval, although only its inequality will be needed.

Let $\Phi(A, B) = (A - B) \log(A/B)$ with its nonnegative extended values. It is jointly convex and lower semicontinuous: it is the sum of the two convex perspectives $A \log(A/B)$ and $B \log(B/A)$. On every bounded collision set, (6.9) and its primed version therefore give lower semicontinuity of the integral of $|a| \Phi$. One can absorb the weight into the arguments because $\Phi(cA, cB) = c\Phi(A, B)$ for $c \geq 0$. The lower semicontinuity follows, for example, by expressing a nonnegative convex lower-semicontinuous function as the supremum of its affine supporting functions and using weak convergence on a finite partition. Exhausting bounded velocity sets, and using that the cutoff is eventually absent on each of them, gives for every $t \in [0, T]$

$$\int_0^t \mathcal{D}(F(s)) \, ds \leq \liminf_N \int_0^t \mathcal{D}_{b_N}(F_N(s)) \, ds.$$

The same convex lower-semicontinuity argument applies to $\mathcal{H}_M(F_N(t))$ under weak L^1 convergence at that time. Combining the two nonnegative terms, using (6.3) and initial entropy convergence, proves

$$\mathcal{H}_M(F(t)) + \int_0^t \mathcal{D}(F(s)) \, ds \leq \mathcal{H}_M(F_0) \quad (0 \leq t \leq T). \quad (6.11)$$

There is no time-dependent subsequence in this conclusion.

In fact the time representative is strongly continuous. In free coordinates the distributional equation and its initial trace read

$$\partial_t [F(t, x + tv, v)] = Q(F, F)(t, x + tv, v).$$

The right side is integrable in (t, x, v) . Subtracting its Bochner integral leaves a distribution with zero time derivative, whose initial value is F_0 . Therefore

$$F(t, x + tv, v) = F_0(x, v) + \int_0^t Q(F, F)(s, x + sv, v) \, ds \quad (6.12)$$

as an $L^1_{x,v}$ identity, initially for almost every time and then for every time by the representative just constructed. The right side is absolutely continuous in L^1 . Free transport is a strongly continuous group on $L^1(\mathbb{T}^3 \times \mathbb{R}^3)$, as follows by approximating an L^1 function by continuous functions with compact velocity support. Thus $F \in C([0, T]; L^1)$. This proves in particular continuity against every bounded measurable test.

Equation (6.12) also holds on almost every individual free path. Since $M(v) > 0$ is constant on such a path, the ordinary absolutely continuous chain rule gives, for every β in Definition 2.1,

$$D_t [M\beta(F/M)] = \beta'(F/M) Q(F, F).$$

The source is integrable because $|\beta'| \leq C_\beta$ and $Q \in L^1$. Moreover

$$|M\beta(f/M) - M\beta(g/M)| \leq C_\beta |f - g|, \quad |M\beta(f/M)| \leq C_\beta f.$$

Hence the renormalization is strongly continuous at the initial and terminal traces. Integrating its path equation proves exactly the renormalized identity and trace in Definition 2.1. The separate integrability requirement follows from $0 \leq Q^\pm / (1 + F/M) \leq Q^\pm$.

Integrating the unrenormalized equation in velocity is justified by absolute collision integrability. The collision integral has zero mass, and the flux is integrable, so it gives the required local continuity equation with its initial density. Momentum has already passed at every time. Finally, the negative part of $F \log F$ is bounded uniformly by mass and energy. For example split $\{F < 1\}$ at $F = e^{-1-|v|^2}$: above that threshold use $-F \log F \leq (1 + |v|^2)F$, and below it use a fixed multiple of $e^{-(1+|v|^2)/2}$. Equation (6.11) then bounds the positive part as well. This proves the uniform absolute logarithmic integral required in the definition.

6.6 Continuation and inherited bounds

At time T , the density just constructed has finite mass, energy, and absolute logarithmic integral. We use the global existence theorem of DiPerna and Lions [3, 4], in the precise periodic formulation of [11, Section 4.1, Theorem 4.1]. We verify its interface with Definition 2.1 before concatenating.

For our kernel $b_{\text{phys}}(\omega, v - w) = |(v - w) \cdot \omega|$, local integrability holds and, for every compact $K \subset \mathbb{R}^3$,

$$\frac{1}{1 + |v|^2} \int_{K \times S^2} b_{\text{phys}}(\omega, v - w) \, d\omega \, dw = \frac{2\pi}{1 + |v|^2} \int_K |v - w| \, dw \longrightarrow 0 \quad (|v| \rightarrow \infty).$$

This is the hypothesis of that theorem. To match its collision and time normalizations, put

$$C_b = \int_{\mathbb{R}^3 \times \mathbb{R}^3 \times S^2} b_{\text{phys}}(\omega, v - w) M(v) M(w) \, d\omega \, dv \, dw, \quad \epsilon = C_b^{-1}.$$

The constant C_b is finite and positive. Use the source kernel b_{phys}/C_b and its equation $\epsilon \partial_s G + v \cdot \nabla_x G = \epsilon^{-1} Q_{b_{\text{phys}}/C_b}(G, G)$, with initial density $F(T)/M$. Here the relative-density operator satisfies $M Q_{b_{\text{phys}}/C_b}(G, G) = Q(MG, MG)/C_b$. Then $\tilde{F}(t, x, v) = M(v)G(\epsilon t, x, v)$ has precisely our collision normalization. The same time substitution converts the source dissipation coefficient ϵ^{-2} into the coefficient one of $\mathcal{D}$.

The cited theorem allows arbitrary finite relative-entropy initial data. It gives weak continuity of G in $L^1(M \, dx \, dv)$, which is exactly weak continuity of $\tilde{F}$ in $L^1(dx \, dv)$, and the renormalized identities with the initial trace for continuously differentiable functions satisfying $|\beta'(z)| \leq C/\sqrt{1+z}$. Our renormalizations are included. Its local mass identity is the one required here. Integrating its momentum identity over the torus removes the spatial divergence of the momentum defect, giving total momentum conservation. This extends to every time by weak continuity and the uniform second-moment bound: the first-moment tail outside $|v| \leq R$ is at most R^{-1} times the second moment. Dropping the nonnegative defect contributions in the theorem's energy and relative-entropy inequalities gives the two inequalities in Definition 2.1 at every time. The negative-entropy estimate above then supplies the uniform absolute logarithmic integral.

We also need separate global integrability of the renormalized gain and loss, rather than only their difference. For any density $f \geq 0$, write $g = f/M$. Since $f/(1+g) \leq M$,

$$\int_{\Omega} \frac{Q^-(f, f)}{1+g} \leq \int_{\Omega} M \nu_f \leq C \int_{\Omega} (1+|v|) f.$$

For $A, B \geq 0$, with the same extended-value convention as for the dissipation,

$$A \leq 2B + \frac{2}{\log 2} (A - B) \log(A/B).$$

Indeed the first term suffices when $A \leq 2B$; otherwise $A - B > A/2$ and $\log(A/B) > \log 2$. Apply this with $A = f' f'_*$, $B = f f_*$, divide by $1 + g(v)$, and integrate with the collision kernel. Nonnegativity of the dissipation gives

$$\int_{\Omega} \frac{Q^+(f, f)}{1+g} \leq 2 \int_{\Omega} \frac{Q^-(f, f)}{1+g} + \frac{8}{\log 2} \mathcal{D}(f).$$

The moment and entropy-dissipation bounds for $\tilde{F}$ make both terms integrable over every finite time interval. Thus the continuation satisfies all of Definition 2.1.

Concatenate $\tilde{F}$ with F at time T . The two weak L^1 traces agree, so the concatenation is continuous against every bounded measurable test. Its renormalized identity follows by adding the two identities: the early terminal term $M\beta(F(T)/M)$ cancels the continuation's

prescribed initial term. The Lipschitz estimate above supplies that early terminal trace. The mass continuity identities concatenate in the same way.

For $t \geq T$, add the continuation inequality

$$\mathcal{H}_M(\tilde{F}(t - T)) + \int_0^{t-T} \mathcal{D}(\tilde{F}(s)) \, ds \leq \mathcal{H}_M(F(T))$$

to (6.11) at T . This gives the global entropy inequality, with no entropy equality at the splice assumed. Energy inequalities and momentum conservation concatenate using the common terminal moments. Collision integrability in the renormalized sense holds on both pieces, hence on every bounded interval of the concatenation. The uniform moment and absolute entropy bounds on such an interval follow from those of the two pieces.

For a dormant sequence, sum the cold upper bounds of Proposition 4.4 over all labels and add $t^2 H$. This is a common, locally integrable pointwise upper bound for the finite total densities. Testing against nonnegative functions and passing to the weak limit preserves it. All assertions of Proposition 6.1 are now proved.

7 The cap dilation and its linear limit

We now let the physical horizons tend to zero. Choose cap shots with $T_N \downarrow 0$, terminal annular scales satisfying $s_N/T_N \rightarrow 0$, and cutoffs $b_N \rightarrow \infty$. Relabeling a subsequence in this way does not require that every integer index occur. Let $t_{*,N} \in [0, T_N]$ be a time attaining the cap. In the periodic lift about the origin define

$$\begin{aligned} r_N &= t_{*,N} + s_N, & h_{r_N}(t, x, v) &= r_N^{-1} h_N(r_N t, r_N x, v), & C_{r_N}(t, x, v) &= r_N c_N(r_N t, r_N x, v), \\ l_N &= \frac{s_N}{r_N} \longrightarrow l \in [0, 1], & U_N &= \frac{T_N}{r_N} \longrightarrow U \in [1, \infty]. \end{aligned} \tag{7.1}$$

The limits are taken after subselection, using the compactification of $[0, \infty)$ for the second ratio. We have $r_N \rightarrow 0$, and $U \geq 1$ follows from $U_N \geq (1 + s_N/T_N)^{-1}$. The prelimit inequality $U_N \geq 1$ is not needed. The dilated cap event occurs at time $1 - l_N$, and

$$0 \leq h_{r_N}(t, x, v) \leq (t + l_N) H(v), \quad 0 \leq t \leq U_N. \tag{7.2}$$

When convenient we suppress the index on r_N .

7.1 The limiting mesh and background

The dilated annular scales are $\mu_{N,j} = s_j/r_N$, $j \leq N$. Their ratio is e^η , their smallest member is l_N , and their largest member tends to infinity. Subselect their logarithmic phase modulo η . The grids then converge on compact subintervals of $(0, \infty)$ to a geometric mesh $\mathcal{M}$ unbounded above. If $l > 0$, its members are $l, e^\eta l, e^{2\eta} l, \dots$. If $l = 0$, it is a two-sided geometric mesh with no smallest member. To verify this description, choose a grid point in $(e^{-\eta}, 1]$, pass to its limit, and multiply it by the fixed factors $e^{j\eta}$. When the smallest point stays positive its integer distance in the grid from that chosen point is bounded, so can also be fixed by subselection. Coincident phase endpoints give the same mesh.

For each $\mu > 0$ and $\sigma \in \{-1, 1\}$, let $u_{\mu,\sigma}(t, z)$ be the unique solution of

$$u = \sigma + \alpha \tanh \frac{\gamma(z - tu)}{\mu}.$$

The left side minus the right side has derivative in u equal to $1 + t\alpha\gamma S(\gamma(z - tu)/\mu)/\mu > 0$; its values have opposite signs for sufficiently large positive and negative u . This proves existence and uniqueness. The implicit function theorem gives smoothness, and

$$|u_{\mu,\sigma} - \sigma| \leq \alpha, \quad \partial_z u_{\mu,\sigma} = \frac{\alpha\gamma S(\gamma(z - tu)/\mu)}{\mu + t\alpha\gamma S(\gamma(z - tu)/\mu)} \geq 0.$$

On an open mesh annulus $\mu < |y| < e^\eta \mu$, define the nonnegative velocity measure

$$\begin{aligned} B(t, x, dp) &= \sum_{\sigma=\pm 1} b_{\mu,\sigma}(t, z) \delta_{u_{\mu,\sigma}(t,z)e_3}(dp), \\ b_{\mu,\sigma}(t, z) &= \frac{\theta K S(\gamma(z - tu_{\mu,\sigma})/\mu)}{\mu + t\alpha\gamma S(\gamma(z - tu_{\mu,\sigma})/\mu)}. \end{aligned} \tag{7.3}$$

Set $B = 0$ in the hole $|y| < l$ if $l > 0$, and assign arbitrary values obeying the same bounds on annular interfaces and the transverse axis. These exceptional positions have zero Lebesgue measure. In particular, for every $k \geq 0$,

$$\int \langle p \rangle^k B(t, x, dp) \leq \frac{C_k}{t + \mu} \leq \frac{C_k}{t} \quad (t > 0) \tag{7.4}$$

on an active annulus. If $l > 0$, the bound is instead uniformly $C_k/(t + l)$ for all $t \geq 0$.

Proposition 7.1. *Every sequence (7.1) has a subsequence for which $h_{r_N} \rightharpoonup g$ in local spacetime L^1 on $(0, U) \times \mathbb{R}_x^3 \times \mathbb{R}_v^3$, where*

$$0 \leq g(t, x, v) \leq (t + l)H(v), \quad D_t g = 2P(B, g) - g\nu_B. \tag{7.5}$$

Here B is (7.3); its gain and loss are defined by integration against the velocity measure. Both terms on the right are locally integrable, with a velocity majorant uniform in position on each compact positive-time interval. Compactly supported velocity averages of h_{r_N} converge strongly in local $L^1_{t,x}$. The limit has a locally strongly continuous L^1 representative at positive times and obeys the integrating-factor formula (7.15) below.

If $l > 0$, pass also to a limit $\lambda_N \rightarrow \lambda_0$ of the shot parameters. The conclusion and its local continuity extend to time zero, with

$$g(0, x, v) = \lambda_0 l \chi(x/l, v). \tag{7.6}$$

7.2 Cold convergence: survival, Jacobian, and stripe averaging

We first identify the cold measure independently of the hot limit. Fix a compact time-space region

$$a \leq t \leq A < U, \quad |x| \leq R,$$

where $a > 0$. Since $U_N \rightarrow U$ and $A < U$, eventually $r_N A < T_N$. The cap therefore holds on the entire initial physical interval $[0, r_N A]$, as required for the survival estimate below. By Proposition 4.4, the dilated cold velocity moments are bounded there, and indeed uniformly in position, by

$$\int \langle p \rangle^k C_{r_N}(t, x, p) dp \leq C_k \left(\frac{1}{t + l_N} + 1 \right). \tag{7.7}$$

The escaped tails have the still smaller bound $C_M r_N^{M+1} t^M e^{-4|p|^2}$ for any fixed $M > 0$. Uniformly decreasing velocity tails hold under these moment bounds.

First restrict to annuli with $\mu_{N,j}$ in a fixed compact interval $[\varepsilon, R_1] \subset (0, \infty)$. There are only finitely many of them, uniformly in N . Apply Corollary 4.5 at physical time $r_N t$ to a color of scale $s = r_N \mu_{N,j}$. Its optical depth is bounded, uniformly for $0 \leq t \leq A$, by $C_A(r_N d_s/s + r_N^2 + r_N s_N) + C_{A,M} r_N^{M+1}$. Here

$$\frac{r_N d_s}{s} = d_0 r_N^{10} \mu_{N,j}^9, \quad r_N^2 + r_N s_N \leq 2r_N^2,$$

so the corollary gives the same-color comparison

$$e^{-\epsilon_N} c_{i,\text{free}} \leq c_i \leq c_{i,\text{free}} + C(d_s/s) m_i, \quad \epsilon_N \longrightarrow 0$$

uniformly on these scales. The lower bound covers the full free core because its velocity parameter lies in the compact support of J . After multiplication by r_N , the remainder has velocity mass

$$r_N O(d_s/s^2) = O(d_0 r_N^9 \mu_{N,j}^8). \quad (7.8)$$

The core moment bound and this estimate show that the dilated actual color and its free core differ by $o(1)$ in velocity total variation, uniformly on compact regions of these annuli. The same holds with each fixed polynomial velocity weight, using compact support of the free-core parameter and the Gaussian remainder.

We compute the free core exactly. Fix one sign and one such annulus, write $\mu_N = s/r_N$, and use the velocity parameters

$$\bar{p} = d_s \bar{w}, \quad p_3 = \sigma + \alpha \tanh \frac{\gamma(z - tp_3)}{\mu_N} + d_s w_3.$$

The axial map depends only on w_3 . Its derivative is

$$\frac{\partial p_3}{\partial w_3} = \frac{d_s}{1 + t\alpha\gamma S(\gamma(z - tp_3)/\mu_N)/\mu_N}.$$

Thus Equation (4.16), after dilation, says that the velocity integral of the free core against a test ψ is

$$\begin{aligned} \int r_N c_{i,\text{free}}(r_N t, r_N x, p) \psi(p) dp &= \int J(w) \frac{KS(\gamma(z - tp_3)/\mu_N)}{\mu_N + t\alpha\gamma S(\gamma(z - tp_3)/\mu_N)} \psi(d_s \bar{w}, p_3) \\ &\quad \cdot \mathbf{1}_E(r_N y - r_N t d_s \bar{w}) \mathbf{1}_{\{|r_N(z - tp_3)| < 1/8\}} dw. \end{aligned} \quad (7.9)$$

On the compact regions under consideration the last axial indicator equals one for all sufficiently large N . Since the implicit velocity map has derivative at least one in p_3 ,

$$|p_3 - u_{\mu_N, \sigma}(t, z)| \leq d_s |w_3|.$$

After the mesh subselection, $\mu_N \rightarrow \mu > 0$, so the velocity test and the smooth amplitude in (7.9) converge uniformly for $w \in \text{supp } J$ to

$$\psi(u_{\mu, \sigma} e_3) \frac{KS(\gamma(z - tu_{\mu, \sigma})/\mu)}{\mu + t\alpha\gamma S(\gamma(z - tu_{\mu, \sigma})/\mu)}.$$

It remains to average the transverse indicator. In dilated coordinates its period is

$$\frac{s^2}{r_N} = r_N \mu_N^2 \longrightarrow 0.$$

The transverse shift of its argument is $td_s \bar{w}$, whose ratio to that period is at most

$$\frac{Ad_s}{r_N \mu_N^2} = Ad_0 r_N^9 \mu_N^8 \longrightarrow 0.$$

For the indicator of the fixed periodic stripe pattern, a translation by this vanishing relative amount changes its integral on a bounded set by $o(1)$, uniformly in its phase and in bounded t, w . One can verify this by summing the lengths of the small intervals around stripe endpoints. The annular margins have dilated width $O(s^2/r_N) = O(r_N \mu_N^2)$, so their area also tends to zero on the annulus. The outer annular radii converge; their boundaries can likewise be removed in sets of arbitrarily small area.

Finally the periodic stripe indicators converge weak-* to their mean θ . Explicitly, if q is the one-period indicator and $q - \theta$ has bounded periodic primitive Q , then for a smooth compactly supported one-dimensional test f , period p , and arbitrary phase ζ ,

$$\left| \int f(y_1)(q(y_1/p + \zeta) - \theta) dy_1 \right| \leq p \|Q\|_\infty \|f'\|_1.$$

Fubini's theorem gives the same assertion with the other coordinates and parameters present. Approximation in L^1 extends it to the annular restrictions. Integrating J , whose mass is one, in (7.9) produces exactly the coefficient $b_{\mu, \sigma}$ in (7.3).

These arguments identify the cold limit on all compact positive scale annuli. To remove the lower scale restriction, use (7.7): on the slab $t \geq a$, the integral over $|y| < C\varepsilon$, with bounded z , is at most $C_{a, A, R, k} \varepsilon^2$. The same estimate holds for the candidate B . Scales larger than a fixed multiple of R have no main color in the chosen spatial compact set, and their escaped tails vanish. We have proved that for every continuous compactly supported velocity test ψ ,

$$\int C_{r_N}(t, x, p) \psi(p) dp \xrightarrow{*} \int \psi(p) B(t, x, dp) \quad \text{in local } L_{t, x}^\infty \quad (7.10)$$

on $(0, U) \times \mathbb{R}^3$. Indeed the computation first gives weak convergence against smooth compact time-space tests, and the common L^∞ bound extends it to all L^1 tests. Polynomial velocity tails can also be removed uniformly. If $l > 0$, choose $\varepsilon < l/2$; all the bounds used above hold down to time zero, so (7.10) does too.

7.3 The hot equation and the size of every error

Under (7.1), the derivative of $r_N^{-1} h_N(r_N t, r_N x, v)$ is the old transport derivative of h_N . Therefore the rescaled equation is exactly

$$D_t h_{r_N} = 2P^{b_N}(C_{r_N}, h_{r_N}) - h_{r_N} \nu_{C_{r_N}}^{b_N} + E_N, \quad (7.11)$$

where

$$\begin{aligned} E_N &= r_N^2 [P^{b_N}(h_{r_N}, h_{r_N}) - h_{r_N} \nu_{h_{r_N}}^{b_N}] + S_N, \\ S_N(t, x, v) &= \sum_{i \neq j} P^{b_N}(c_{i, N}, c_{j, N})(r_N t, r_N x, v). \end{aligned}$$

There is no additional factor in S_N . Its smallness comes from the physical mixed-cold source estimate: for every sufficiently large fixed M ,

$$\begin{aligned} 0 &\leq S_N(t, x, v) \leq C_M r_N^{M-1} t^{M-1} H(v), \\ r_N^2 P^{b_N}(h_{r_N}, h_{r_N}) &\leq C r_N^2 (t + l_N)^2 H(v), \\ r_N^2 h_{r_N} \nu_{h_{r_N}}^{b_N} &\leq C r_N^2 (t + l_N)^2 \langle v \rangle H(v). \end{aligned} \quad (7.12)$$

The second line uses Lemma 3.7; the last line uses the first velocity moment of H . Thus on every fixed compact time interval, $E_N \rightarrow 0$ in local spacetime L^1 , with an error bounded by a constant times $r_N^2 + r_N^{M-1}$ after integration over a fixed spatial compact set. Choose $M > 1$.

The main cold sheets and Lemma 3.4 also give

$$P^{b_N}(C_{r_N}, H) \leq C \left(\frac{1}{t + l_N} + 1 \right) H, \quad \nu_{C_{r_N}}^{b_N}(v) \leq C \left(\frac{1}{t + l_N} + 1 \right) \langle v \rangle.$$

The escaped cold component is bounded by a vanishing multiple of $e^{-4|v|^2} \leq CH$, so its gain is covered by Lemma 3.7. Equations (7.2) and (7.11) now imply, on $a \leq t \leq A < U$,

$$|D_t h_{r_N}(t, x, v)| \leq C_{a,A} \langle v \rangle H(v). \quad (7.13)$$

This is a genuine integrable velocity majorant, uniform in position. If $l > 0$, the same conclusion holds on $0 \leq t \leq A$, since $l_N \geq l/2$ eventually.

The cap gives weak compactness of h_{r_N} in local L^1 , and (7.13) permits application of Lemma 6.2. Subselect on an exhaustion of $(0, U) \times \mathbb{R}^3$. The resulting limit g satisfies the bound in (7.5), and each compactly supported velocity average converges strongly in local $L_{t,x}^1$.

7.4 Passing the cold-hot products

The dilated cold densities concentrate in velocity, so the two-density compactness argument of Section 6 does not apply to C_{r_N} . The needed product passage uses (7.10) and hot averaging instead.

For compact velocity tests ψ, χ , put

$$a_N(t, x) = \int \psi(p) C_{r_N}(t, x, p) dp, \quad \zeta_N(t, x) = \int \chi(v) h_{r_N}(t, x, v) dv.$$

On every compact positive-time region, a_N is bounded in L^∞ and converges weak-* to $a = \int \psi dB$, while $\zeta_N \rightarrow \zeta = \int \chi g dv$ strongly in L^1 . For a bounded time-space test φ ,

$$\int \varphi(a_N \zeta_N - a \zeta) = \int \varphi a_N (\zeta_N - \zeta) + \int \varphi (a_N - a) \zeta \rightarrow 0.$$

Finite sums of separated velocity tests approximate continuous tests uniformly on compact velocity products. The total product mass is uniformly bounded on compact time-space regions by (7.7) and the cap. We conclude

$$C_{r_N}(t, x, p) h_{r_N}(t, x, v) dp dv dx dt \rightharpoonup B(t, x, dp) g(t, x, v) dv dx dt \quad (7.14)$$

against continuous compact velocity tests, with arbitrary smooth compact time-space dependence. Uniform moment bounds in both velocities allow removal of velocity cutoffs in tests with any fixed polynomial growth for which they are used below.

Here is an explicit weak collision test suitable for this convergence. For a continuous output test ϕ , let

$$\mathcal{K}_\phi(p, v) = \frac{|p - v|}{2} \int_{S^2} \phi \left(\frac{p + v}{2} + \frac{|p - v|}{2} \omega \right) d\omega.$$

Lemma 3.1 gives $\int \phi P(c, h) = \iint c(p) h(v) \mathcal{K}_\phi(p, v) dp dv$. The test $\mathcal{K}_\phi$ is continuous in both incoming velocities, including $p = v$, and is bounded in absolute value by $2\pi|p - v|\|\phi\|_\infty$. The loss test is $2\pi|p - v|\phi(v)$, with the same properties. Inserting smooth time-space dependence in ϕ does not change these facts. On bounded velocity products the factor $\mathbf{1}_{\{|p-v|<b_N\}}$ is eventually one; its complement is uniformly negligible by the weighted product tails. Applying (7.14) proves the required gain and loss limits.

The limiting gain is an integrable function, rather than an additional singular output measure. Indeed each summand of B is an allowed zero-width axial sheet in Lemma 3.4; hence on $a \leq t \leq A$,

$$0 \leq P(B, g) \leq C_{a,A} H, \quad 0 \leq g\nu_B \leq C_{a,A} \langle v \rangle H.$$

Equivalently the weak gain measure is dominated by this integrable function, which also verifies absolute continuity at the velocity axis. Passing (7.11), using (7.12), proves the equation in (7.5).

7.5 A positive lower mesh endpoint and the local mild formula

When $l > 0$, (7.13) holds uniformly down to zero. The rescaled initial hot density, on every fixed spatial compact set for large N , is exactly

$$h_{r_N}(0, x, v) = \lambda_N l_N \chi(x/l_N, v).$$

The other periodic copies have moved outside that compact set. Subselect $\lambda_N \rightarrow \lambda_0$. Smoothness and compact support of χ , and $l_N \rightarrow l > 0$, give strong local L^1 convergence to (7.6). Passing the weak transport identity with tests touching time zero is justified by the uniform source majorant and yields that initial trace for g . The cold product argument also extends to zero in this case, as already noted. For $l = 0$ the coefficient can be of order $1/t$, and the proposition only asserts the equation on the open positive-time interval.

On any $[a, A] \subset (0, U)$, the limiting transport source is locally integrable and has the preceding velocity majorant uniformly in position. Changing to free coordinates as in (6.12), first on bounded velocities and enlarged spatial compact sets, gives a locally strongly continuous L^1 representative. The uniform integrable velocity tails then remove the velocity truncation. The cap bound passes to every trace of this representative. Almost every free path obeys the scalar absolutely continuous equation $D_t g + \nu_B g = 2P(B, g)$. Its loss coefficient is locally integrable along the path, since $\nu_B(t, x, v) \leq C_a \langle v \rangle$ for $t \geq a$. The integrating factor therefore gives, for $0 < a < t < U$,

$$\begin{aligned} g(t, x, v) &= g(a, x - (t - a)v, v) \exp \left[- \int_a^t \nu_B(\tau, x - (t - \tau)v, v) d\tau \right] \\ &\quad + 2 \int_a^t P(B, g)(s, x - (t - s)v, v) \exp \left[- \int_s^t \nu_B(\tau, x - (t - \tau)v, v) d\tau \right] ds. \end{aligned} \tag{7.15}$$

This is an almost-everywhere identity, or equivalently an identity in local L^1 at the displayed times. It holds with $a = 0$ when $l > 0$. Altering the medium on the null annular interfaces does not change the identity: free transport preserves phase-space measure, so Fubini's theorem discards the affected set of paths. These statements complete the proof of Proposition 7.1.

8 A cap event produces a nonzero limit

The weak limit in Proposition 7.1 is obtained on an open time interval. A supremum attained by an approximation could in principle disappear through concentration, or through escape from that interval. The purpose of this section is to exclude both possibilities. All constants used below may depend on the fixed jet parameters. None depends on the approximation index.

Proposition 8.1 (Nonvanishing of every cap limit). *Consider any sequence of cap shots with horizons $T_n \rightarrow 0$, approximation indices $N_n \rightarrow \infty$, and $s_{N_n}/T_n \rightarrow 0$. Make the rescaling (7.1), and pass to any subsequence supplied by Proposition 7.1. Its limit satisfies*

$$g \not\equiv 0 \quad \text{on } (0, U) \times \mathbb{R}_x^3 \times \mathbb{R}_v^3.$$

This conclusion includes $l = 1$, $U = 1$, and $U = \infty$.

We first describe the positive kernels used in the proof. Subsequently we establish the two different mechanisms needed near and away from the velocity axis.

8.1 A positive majorizing equation and its kernels

For this section relabel the sequence by n , set $U_n = T_n/r_n$, and write

$$r_n = t_{*,n} + s_{N_n}, \quad \ell_n = \frac{s_{N_n}}{r_n}, \quad \tau_n = \frac{t_{*,n}}{r_n} = 1 - \ell_n, \quad h_n = h_{r_n}, \quad C_n = C_{r_n}, \quad f_n = \frac{h_n}{H}.$$

Then $\ell_n \rightarrow l$. In particular $0 \leq \ell_n \leq 1$, $0 \leq \tau_n \leq 1$. The cap gives, on $0 \leq t \leq U_n$,

$$0 \leq f_n(t, x, v) \leq t + \ell_n, \quad \|f_n(\tau_n)\|_\infty = 1. \quad (8.1)$$

Essential supremum norms and pointwise formulas involving H always ignore the velocity axis, a null set. The mild representatives of the finite approximations will be used at the cap times. Choose Borel representatives when composing densities with the maps below, enforcing their asserted bounds on exceptional null sets as well. Characteristic identities and endpoint choices are still made only on their common full-measure sets.

Lemma 8.2 (Core kernels on the dilated scale). *There are nonnegative velocity measures $A_n(t, x, dp)$ and numbers $\varepsilon_n \rightarrow 0$ with the following properties. After discarding finitely many terms so that $r_n \leq \bar{T}$, the measures A_n are defined from free transport on the whole interval $0 \leq t \leq 1$, independently of the cap shot's existence interval. The bounds on A_n below hold on that whole interval; comparison with h_n is used only while the shot exists.*

(i) On $0 < t < \min(1, U_n)$, along almost every free characteristic,

$$D_t h_n \leq 2P(A_n, h_n) + \varepsilon_n H. \quad (8.2)$$

At each point A_n contains at most one main free core. Its dilated scale $\mu = s/r_n$ satisfies $|y| \asymp \mu$, with coordinates measured from its periodic copy, and

$$\frac{2P(A_n, H)}{H} \leq \frac{C}{t + \ell_n}.$$

(ii) Before spatial masks are imposed, a core of scale $\mu = s/r_n$ and sign σ is the pushforward of

$$\frac{KS(\gamma(z - tp_3)/\mu)}{\mu + t\alpha\gamma S(\gamma(z - tp_3)/\mu)} J(w) dw, \quad \bar{p} = d_s \bar{w}, \quad p_3 = \sigma + \alpha \tanh(\gamma(z - tp_3)/\mu) + d_s w_3. \quad (8.3)$$

The actual masks are measurable factors between zero and one. The density in (8.3) is at most $C(t + \mu)^{-1} J(w)$.

(iii) For $t \geq a > 0$, the normalized gain measure

$$\mathcal{K}_n(t, x, v; dp d\xi) = 4A_n(t, x, dp) \delta((v - p) \cdot (\xi - v)) \frac{H(\xi)}{H(v)} d\xi \quad (8.4)$$

has total mass at most C/a . Uniformly in its base point, the contributions from large $|p| + |\xi - v|$, or from large retained Gaussian parameters in a product-sheet majorant, tend to zero. Its total mass tends uniformly to zero as $|v| \rightarrow \infty$. The axial interval, axial mismatch, and small transverse input conclusions of Lemma 3.5 apply to $\mathcal{K}_n$, with the amplitude factor C/a .

Proof. For $0 \leq t \leq 1$, let i be the retained label whose enlargement $\mathcal{O}_i(r_n t)$ contains $r_n x$, if there is one, and define

$$A_n(t, x, dp) = r_n f_i(r_n t, r_n x, p) dp;$$

set $A_n = 0$ when no such label exists. The free cores f_i and enlargements are defined from the datum, so this definition does not require a cap shot beyond U_n . Their disjointness and free-core bounds hold because $r_n t \leq \bar{T}$.

On $0 \leq t \leq \min(1, U_n)$, Proposition 4.4 compares the actual cold density with these free cores. All other free cores belong to the escaped sum. The main remainder has, after dilation, integrated sheet amplitude at most

$$Cr_n \frac{d_s}{s^2} = Cr_n d_0 s^8 \leq Cr_n.$$

The escaped sum is bounded by $C_A r_n (r_n t)^A e^{-4|v|^2}$. Use Lemma 3.4 for the main narrow remainder. For the escaped term, $e^{-4|v|^2} \leq CH(v)$ and Lemma 3.7 give the required gain bound. Together these supply a positive remainder R_n with $C_n \leq A_n + R_n$ and $P(R_n, H)/H = o(1)$, uniformly on that interval. There the cap gives $h_n \leq 2H$.

In the dilated hot equation the hot-hot gain is $r_n^2 P(h_n, h_n)$, bounded by $Cr_n^2 H$ by Lemma 3.7. The mixed cold source is bounded by $C_A (r_n t)^{A-1} H$; choose $A > 1$. Ignore all losses and combine these bounds with the remainder just described. This proves (8.2) with $\varepsilon_n \rightarrow 0$, uniformly down to time zero.

Equation (8.3) is (4.16) after multiplying the cold density by r_n . Since the smallest scale is ℓ_n , its amplitude is bounded by $C/(t + \ell_n)$. The geometric restriction comes from the enlarged transverse set of the main color; its relative enlargement is $O(s)$, uniformly small.

The coefficient 4 in (8.4) is the product of the coefficient 2 in the linear hot gain and the coefficient 2 in Lemma 3.1. The gain and velocity-tail bounds follow from Lemmas 3.4 and 3.5. For parameter truncations, apply Lemma 3.6 with the common bounds $A = 1 + \alpha$, $C_0 = C_J$, and the physical width d_s . The coefficient $C/(t + \mu) \leq C/a$ on this slab remains a domination amplitude outside the normalized parameter law. The parameter-tail estimate controls the complete normalized gain, including $H(\xi)/H(v)$, uniformly under the axial compression in (8.3). For the exact free core, J already has bounded parameter support. $\square$

For a fixed $a > 0$, define the backward Volterra operator

$$(\mathcal{V}_n F)(t, x, v) = \int_a^t \int F(s, x - (t-s)v, \xi) \mathcal{K}_n(s, x - (t-s)v, v; dp d\xi) ds. \quad (8.5)$$

Representatives in the finite-temperature iteration. For each finite n , the core measure has a velocity density, $A_n(s, x, dp) = a_n(s, x, p) dp$. Integrating first in p in (8.4) shows that, for $\xi \neq v$, its input marginal is

$$\mathcal{K}_n(s, x, v; \mathbb{R}^3, d\xi) = \frac{4H(\xi)}{|\xi - v|H(v)} \left[\int_{(v-p) \cdot (\xi-v)=0} a_n(s, x, p) dA_p \right] d\xi.$$

There is no atom at $\xi = v$: each plane-area measure gives that point zero mass, and $p = v$ is null for the cold density. For a null set in (s, X, ξ) , integrate also in the endpoint variables and use the measure-preserving change $X = x - (t-s)v$. Tonelli's theorem then shows that altering the input on that set changes $\mathcal{V}_n F(t)$ only on a null set of endpoints, at each fixed terminal time. Finite repetition preserves this property. A countable sequence of lower times $a \downarrow 0$ suffices below; intersecting the good endpoint sets for these times, the approximation indices, and the finitely many iterations gives the common full-measure sets used at the cap events.

Positivity and Lemma 8.2 give, for a bounded nonnegative F ,

$$\|\mathcal{V}_n^j F(t)\|_\infty \leq \|F\|_\infty \frac{(C \log(t/a))^j}{j!}, \quad a \leq t \leq 1. \quad (8.6)$$

Indeed, the ordered j -time integral of $\prod_{i=1}^j C/t_i$ equals $(C \log(t/a))^j/j!$. In particular every fixed number of further gains has bounded mass on this slab.

8.2 Reduction to a finite gain integral

We begin the proof of Proposition 8.1. The case $l > 0$ is settled directly. For $l = 0$, the cap will force a positive contribution from five successive gains; the remaining subsections show that this contribution vanishes if the weak limit is zero.

Pass to a subsequence on which the shot parameters converge, $\lambda_n \rightarrow \lambda_\infty$. If $l > 0$ and $\lambda_\infty > 0$, Proposition 7.1 gives the initial hot trace

$$g(0, x, v) = \lambda_\infty l \chi(x/l, v).$$

This trace is nonzero. More explicitly, the limiting loss frequency obeys

$$\nu_B(t, x, v) \leq \frac{C(1 + |v|)}{t + l}.$$

On any bounded velocity set and any finite interval beginning at zero its integral along a flight is finite. The positive mild formula for (7.5) therefore retains the initial trace multiplied by a strictly positive survival factor. A bounded positive portion of the smooth nonzero bump gives positive interior mass.

If $l > 0$ and $\lambda_\infty = 0$, use Lemma 8.2 from time zero. The initial ratio has norm at most $\lambda_n \ell_n \|\chi/H\|_\infty$, independently of the spatial scaling of the bump. On $0 \leq t \leq \tau_n \leq 1$, eventually $t + \ell_n \geq l/2$. The positive integral inequality and the elementary scalar integrating-factor estimate give

$$\sup_{0 \leq t \leq \tau_n} \|f_n(t)\|_\infty \leq (\lambda_n \ell_n \|\chi/H\|_\infty + \varepsilon_n) e^{2C/l} \rightarrow 0.$$

This contradicts (8.1). Both alternatives also apply when $l = 1$, even if $\tau_n \rightarrow 0$ or some hit occurs at time zero.

It remains to consider $l = 0$, so $\tau_n \rightarrow 1$. Suppose, to obtain a contradiction, that $g = 0$ throughout the open interval $(0, U)$. Nonnegativity upgrades this zero weak limit to local L^1 convergence: if K is a compact subset of the open spacetime-velocity region, take a nonnegative smooth compact test at least one on K . Its pairing with h_n tends to zero, hence $\int_K h_n \rightarrow 0$. Away from the velocity axis H has a positive lower bound on compact velocity sets. Consequently

$$f_n \longrightarrow 0 \quad \text{in measure on compact subsets of } (0, U) \times \mathbb{R}_x^3 \times \{\bar{v} \neq 0\}. \quad (8.7)$$

Fix $0 < a < 1/4$, so that $a < \tau_n$ for large n . The positive mild inequality (8.2) reads

$$f_n \leq d_n + \mathcal{V}_n f_n + e_n,$$

where $d_n(t, x, v) = f_n(a, x - (t - a)v, v)$, $\|d_n\|_\infty \leq a + \ell_n$, and $0 \leq e_n \leq \varepsilon_n$ on this unit-length interval. Iterate the gain term five times. Using positivity and (8.6) gives

$$f_n(t, x, v) \leq (a + \ell_n + \varepsilon_n) \sum_{j=0}^4 \frac{(C \log(1/a))^j}{j!} + (\mathcal{V}_n^5 f_n)(t, x, v), \quad a \leq t \leq \tau_n. \quad (8.8)$$

Every formula is used on a common full-measure set of characteristics for the finite approximation.

The term outside $\mathcal{V}_n^5 f_n$ becomes arbitrarily small when first $n \rightarrow \infty$ and then $a \downarrow 0$. We therefore need to show that the five-fold term tends to zero at endpoints carrying a fixed positive fraction of the cap. We first control those endpoints, then separate axial and nonaxial limiting velocities.

8.3 Compactness of the relevant endpoints

Lemma 8.3 (Endpoint and path truncations). *Fix $a > 0$, and let $t_n \in [a, 1]$. In a five-fold integral $(\mathcal{V}_n^5 F_n)(t_n, x_n, v_n)$, assume $0 \leq F_n \leq 2$. For every prescribed error, one may restrict all $|p_i| + |v_i - v_{i-1}|$ and sheet parameters to fixed bounded sets at that error, uniformly in n, t_n, x_n, v_n . Moreover:*

- (i) *if $|v_n| \rightarrow \infty$, the whole integral tends to zero;*
- (ii) *after retaining bounded endpoint velocities, an endpoint escaping every bounded neighborhood of every dilated periodic origin also has a vanishing integral.*

Consequently any sequence along which the five-fold integral has a positive lower bound has, after periodic translations and subselection, convergent endpoint positions and velocities.

Proof. At any one gain, a removed normalized tail is uniformly small by Lemma 8.2. Bound the remaining four gains and the input F_n by (8.6). Summing over the five possible locations of a removed tail proves the first assertion. The same argument at the first gain, using its uniform decay for large outgoing v_n , proves (i).

On the retained paths all velocities are bounded by a fixed number, and the sum of free-flight lengths is at most that number. Thus every collision position is a bounded dilated distance from the endpoint. If its transverse distance from every periodic origin tends to infinity, each encountered main scale satisfies $\mu = s/r_n \rightarrow \infty$, or no core is encountered. The bound $C/(t + \mu) \leq C/\mu$ makes the first nonzero core kernel negligible.

Suppose instead that the transverse coordinate is bounded relative to a periodic origin while the axial coordinate escapes all such origins. Encountered main scales satisfy $0 < \mu \leq M$. For bounded p_3, t , their amplitudes obey

$$\frac{KS(\gamma(z - tp_3)/\mu)}{\mu + t\alpha\gamma S(\gamma(z - tp_3)/\mu)} \leq \frac{K}{\mu} S(\gamma(z - tp_3)/\mu) \longrightarrow 0 \quad (8.9)$$

uniformly in $0 < \mu \leq M$ as $|z| \rightarrow \infty$. For example, when $|z - tp_3| \geq Z$, the right side is at most $4K\mu^{-1}e^{-2\gamma Z/\mu}$; its supremum over $0 < \mu \leq M$ tends to zero. The core masks can only decrease these bounds. The same estimates hold for the first gain of every retained path, and the remaining gains have uniformly bounded mass.

The period of the dilated density is $1/r_n \rightarrow \infty$. Choose a nearest periodic origin before extracting the endpoint subsequence. A path of bounded dilated length cannot visit neighborhoods of two different such origins for large n . The two preceding alternatives cover failure of boundedness relative to the chosen origin, and prove (ii). $\square$

8.4 Two gains at an axial endpoint

For $\mu > 0$ and $\sigma \in \{-1, 1\}$, denote by $u_{\mu,\sigma}(t, z)$ the solution of

$$u = \sigma + \alpha \tanh(\gamma(z - tu)/\mu).$$

This notation extends the profile independently of every annular or stripe mask.

Lemma 8.4 (Characteristic matching and small scales). *On every compact set with $t \geq a > 0$, the following statements hold uniformly.*

(i) *If $a \leq t_2 \leq t_1$, $\Delta = t_1 - t_2$, and $z_2 = z_1 - \Delta w$, then*

$$|w - u_{\mu,\sigma}(t_1, z_1)| \leq |w - u_{\mu,\sigma}(t_2, z_2)|. \quad (8.10)$$

(ii) *As $\mu \downarrow 0$, the distance of $u_{\mu,\sigma}(t, z)$ to*

$$\{z/t, \sigma - \alpha, \sigma + \alpha\}$$

tends to zero uniformly. For the perturbed axial velocity in (8.3),

$$|p_3 - u_{\mu,\sigma}(t, z)| \leq d_s |w_3|. \quad (8.11)$$

Proof. The profile satisfies

$$\partial_z u_{\mu,\sigma} = \frac{(\alpha\gamma/\mu)S}{1 + (t\alpha\gamma/\mu)S} \geq 0, \quad \partial_t u_{\mu,\sigma} + u_{\mu,\sigma} \partial_z u_{\mu,\sigma} = 0.$$

For fixed t_1, t_2, z_1 , the function $F(w) = w - u_{\mu,\sigma}(t_2, z_1 - \Delta w)$ has derivative $F'(w) \geq 1$. Its root is $u_{\mu,\sigma}(t_1, z_1)$: the defining implicit equation shows that this velocity is unchanged along the straight segment between the two points. Integrating $F' \geq 1$ proves (8.10).

Fix $0 < \delta < \alpha$. If $u_{\mu,\sigma}$ is within δ of an endpoint speed, there is nothing to prove. Otherwise

$$\left| \frac{u_{\mu,\sigma} - \sigma}{\alpha} \right| \leq 1 - \frac{\delta}{\alpha}, \quad |z - tu_{\mu,\sigma}| \leq \frac{\mu}{\gamma} \operatorname{arctanh}(1 - \delta/\alpha).$$

It follows that

$$|u_{\mu,\sigma} - z/t| \leq \frac{\mu}{\gamma a} \operatorname{arctanh}(1 - \delta/\alpha).$$

First choose δ , then μ small. This proves the uniform assertion in (ii). Finally, the function

$$p \longmapsto p - \sigma - \alpha \tanh(\gamma(z - tp)/\mu)$$

has derivative at least one. Its values at $u_{\mu,\sigma}$ and at the perturbed p_3 are 0 and $d_s w_3$. This proves (8.11). $\square$

Lemma 8.5 (Vanishing of two axial gains). *Fix $a > 0$. Let $t_n \in [a, 1]$, and suppose the endpoint positions and velocities are bounded with $\bar{v}_n \rightarrow 0$. Then*

$$(\mathcal{V}_n^2 1)(t_n, x_n, v_n) \longrightarrow 0.$$

The conclusion continues to hold if all subsequent factors in a longer finite gain integral are bounded uniformly.

Proof. Remove normalized parameter and increment tails at an arbitrarily small cost, using Lemma 8.2. On the retained two-gain paths, all velocities and positions lie in fixed compact sets, the times lie in $[a, 1]$, and every active scale satisfies $0 < \mu \leq M$. In particular the largest possible physical width is $d_0(r_n M)^{10} \rightarrow 0$.

Write $t_1, x_1 = (y_1, z_1)$ for the first collision point, and ξ for its input velocity. The next point has axial coordinate $z_2 = z_1 - (t_1 - t_2)\xi_3$. All estimates below are uniform when t_1, x_1 vary in their retained compact sets. At the first gain, Lemma 3.5 supplies a common modulus $\omega(\rho) \downarrow 0$ for the mass of any axial interval of length ρ ; the interval's center is arbitrary. It also says that for each fixed $b > 0$, the first-kernel mass on $\{|\bar{\xi}| > b\}$ tends to zero as $\bar{v}_n \rightarrow 0$.

Here is an order of choices that handles all scales at the second point. Fix an error $\epsilon > 0$, including the fixed factors from the two time integrations and the bound C/a . Choose $\delta > 0$ so that the sum of the first-kernel masses of the δ -neighborhoods of the five numbers

$$z_1/t_1, \quad 1 - \alpha, \quad 1 + \alpha, \quad -1 - \alpha, \quad -1 + \alpha \tag{8.12}$$

is at most ϵ . This is possible uniformly because there are five intervals and their lengths are 2δ . Lemma 8.4 then gives $\mu_0 > 0$, chosen uniformly on the retained compact sets, so that every profile with $\mu < \mu_0$ is within $\delta/4$ of its corresponding three collapsed speeds.

The geometric mesh has a uniformly bounded number J_0 of scales in $[\mu_0, M]$. Choose $b_0 > 0$ so small that the first-kernel mass of the union of the b_0 -neighborhoods of

$$\{u_{\mu,\sigma}(t_1, z_1) : \mu \in [\mu_0, M] \text{ in the current mesh, } \sigma = \pm 1\}$$

is at most ϵ . For example it suffices that $2J_0\omega(2b_0) \leq \epsilon$, after incorporating the fixed amplitude bound into ω . This list may depend on n ; its cardinality bound does not. Its centers may depend on t_1, z_1 , which are held fixed when the first kernel is integrated. Thus neither dependence weakens the interval bound.

Let $E_n(t_1, z_1)$ be the union of these exceptional intervals. Outside E_n , every retained cold speed at the second point has an axial mismatch bounded below by a fixed positive number for all large n . For $\mu \geq \mu_0$, this follows from (8.10) and (8.11), with lower bound $b_0/2$. For

a constant collapsed speed it follows immediately, with lower bound $\delta/2$. For the remaining branch use the exact identity

$$\xi_3 - \frac{z_2}{t_2} = \frac{t_1}{t_2} \left(\xi_3 - \frac{z_1}{t_1} \right);$$

the factor $t_1/t_2 \geq 1$ preserves the lower bound. The errors $\delta/4$ in small-scale approximation and the vanishing thermal errors can be absorbed. We have therefore obtained, on the complement of E_n ,

$$|\xi_3 - p_3^{(2)}| \geq c_* := \frac{1}{4} \min(\delta, b_0) > 0$$

for every retained second-core parameter.

The second-kernel axial mismatch estimate in Lemma 3.5 now tends uniformly to zero as $|\bar{\xi}| \rightarrow 0$. Choose $b > 0$ so small that this second kernel has mass at most ϵ whenever $|\bar{\xi}| \leq b$ and the displayed mismatch holds. For this fixed b , the first-kernel mass on $|\bar{\xi}| > b$ tends to zero. Its mass on E_n is at most 2ϵ , while the remaining first-kernel mass is bounded. Combining the three regions, then removing the initial tail error, gives a bound tending to zero as $\epsilon \downarrow 0$. All choices preceded the final passage $n \rightarrow \infty$, and were uniform in the two times. This proves the lemma.

In a longer integral, all factors following these two gains are bounded by (8.6); the same proof applies. $\square$

8.5 Finite-dimensional regularization away from the axis

The next two lemmas make explicit the relation between plane area, redundant angular parameters, and the pullback of sets of small spacetime–velocity volume.

Lemma 8.6 (Redundant plane parameters). *Let $n \neq 0$, $e = n/|n|$, and let ζ be a bounded nonnegative function supported in $[-1, 1]$ with integral one. For every nonnegative measurable F ,*

$$\int_{n^\perp} F(k) \frac{dA(k)}{|n|} = \int_{\mathbb{R}^3} F(n \times L) |n| \zeta(L \cdot e) dL. \quad (8.13)$$

If $\delta \leq |n| \leq R$ and $|k| \leq R_1$, the right side is supported in

$$|L| \leq \sqrt{(R_1/\delta)^2 + 1}$$

and has density at most $R\|\zeta\|_\infty$. The identity can be applied successively when each normal is a measurable function of parameters chosen at earlier gains.

Proof. Write $L = L^\perp + \lambda e$. The map $L^\perp \mapsto n \times L^\perp$ multiplies two-dimensional area by $|n|^2$. Integration in λ contributes $\int \zeta = 1$, proving (8.13) and the support and density bounds. For successive gains, fix all previous parameters before applying the identity. The current normal is then fixed; Fubini's theorem for nonnegative functions justifies iteration. No derivatives of the variable normals enter this conditional change of variables. $\square$

Lemma 8.7 (Uniform pullback estimate). *Let $\mathcal{O} \subset \mathbb{R}^k$ be open, $k \geq d$, and suppose $\Phi_n, \Phi : \mathcal{O} \rightarrow \mathbb{R}^d$ satisfy $\Phi_n \rightarrow \Phi$ in C_{loc}^1 and $\text{rank } D\Phi = d$ almost everywhere. Let $K \Subset \mathcal{O}$ and $B \subset \mathbb{R}^d$ be compact. Let a_n be measurable, supported in K , with $|a_n| \leq A$. Let F_n be measurable, supported in B , with $|F_n| \leq M$, and converging to zero in measure. Then*

$$\int_{\mathcal{O}} |F_n(\Phi_n(q)) a_n(q)| dq \longrightarrow 0. \quad (8.14)$$

No convergence or continuity of the densities a_n is required.

Proof. Boundedness and convergence in measure on the finite-volume set B imply $\|F_n\|_{L^1(\mathbb{R}^d)} \rightarrow 0$: for every $\eta > 0$, bound the integral by $\eta|B| + M|\{|F_n| > \eta\}|$.

Fix $\epsilon > 0$. The critical set in K has measure zero, so choose an open neighborhood E of it with $|E \cap K| < \epsilon$. The compact set $K \setminus E$ consists of regular points. At such a point q_0 , choose d input coordinates giving a nonzero minor of $D\Phi(q_0)$. Let π be projection onto the complementary $k - d$ coordinates, and define

$$G(q) = (\Phi(q), \pi q), \quad G_n(q) = (\Phi_n(q), \pi q), \quad A_0 = DG(q_0).$$

The matrix A_0 is invertible. Choose a convex ball $V \Subset \mathcal{O}$ about q_0 so small that, for all sufficiently large n ,

$$\sup_{q \in V} \|A_0^{-1} DG_n(q) - I\| < \frac{1}{2}.$$

Integration on the line segment between $q, q' \in V$ gives

$$|A_0^{-1}(G_n(q) - G_n(q')) - (q - q')| \leq \frac{1}{2}|q - q'|.$$

Thus G_n is injective on V , and its Jacobian has a positive lower bound c_V independent of n . The inverse function theorem consequently makes it a diffeomorphism onto its image.

Cover $K \setminus E$ by finitely many such balls V_j . The change-of-variables formula gives

$$\begin{aligned} \int_{V_j \cap K} |F_n(\Phi_n(q))| dq &\leq c_{V_j}^{-1} \int_{G_{n,j}(V_j \cap K)} |F_n(z)| dz dw \\ &\leq c_{V_j}^{-1} |\pi_j(V_j)| \|F_n\|_{L^1(\mathbb{R}^d)}. \end{aligned}$$

When $k = d$, the zero-dimensional factor has value one. The integral on $E \cap K$, including its density, is at most $AM\epsilon$. Sum over the finite cover, pass to the limit in n , and then let $\epsilon \downarrow 0$. This proves (8.14). $\square$

Lemma 8.8 (Five-gain rank and robust pullback). *Fix a nonaxial endpoint velocity v_0 , an endpoint (t_0, x_0) with $t_0 > 0$, five positive scales μ_i , and five signs σ_i . Given positive times t_i and $L_i \in \mathbb{R}^3$, define recursively*

$$\begin{aligned} x_i &= x_{i-1} - (t_{i-1} - t_i)v_{i-1}, \\ p_i &= u_{\mu_i, \sigma_i}(t_i, z_i)e_3, \quad n_i = v_{i-1} - p_i, \\ k_i &= n_i \times L_i, \quad v_i = v_{i-1} + k_i, \quad i = 1, \dots, 5. \end{aligned} \tag{8.15}$$

The map

$$\Phi : (t_1, \dots, t_5, L_1, \dots, L_5) \mapsto (t_5, x_5, v_5)$$

is real analytic on the connected open set $(0, \infty)^5 \times \mathbb{R}^{15}$, and has rank seven almost everywhere. In particular it is a submersion almost everywhere on the ordered region $0 < t_5 < \dots < t_1 < t_0$.

For convergent endpoint sequences with a nonaxial limiting velocity, convergent positive scale sequences, and the finite-temperature core formulas (8.3), the analogous maps converge in C^1 on compact parameter sets to Φ . Bounded measurable parameter densities supported in a common compact set therefore satisfy Lemma 8.7, including after any measurable spatial, velocity, or time restrictions.

Proof. For $t > 0$, the derivative with respect to u in the defining implicit equation is

$$1 + \frac{t\alpha\gamma}{\mu} S(\gamma(z - tu)/\mu) > 0.$$

The implicit analytic function theorem gives an analytic profile for every $z \in \mathbb{R}$. Uniqueness, which also follows from this strict monotonicity, identifies the local profiles on overlaps. The recursive formulas (8.15) are consequently analytic. They make sense even without ordering the positive times: a negative formal free-flight duration merely changes the point at which the next analytic profile is evaluated. This supplies the connected open domain stated in the lemma.

We exhibit one rank-seven point. Fix $t_5 = \tau$ with $0 < \tau < t_0$, and set all five collision times equal to τ . After a rotation about the velocity axis, write $v_0 = (R, 0, z)$, $R > 0$. All collision positions now coincide at $x_0 - (t_0 - \tau)v_0$, independently of the angular parameters. Thus write the five axial velocities simply as $p_i = a_i e_3$; their values a_i may be arbitrary.

Choose

$$k_1 = (0, 1, 0), \quad k_2 = (1, -R, 0).$$

They satisfy $k_1 \perp n_1$, $k_2 \perp n_2$, and span the transverse plane. Put

$$w = \bar{v}_2 = (R + 1, 1 - R) \neq 0, \quad b_3 = v_{2,3} - a_3,$$

and choose a sufficiently small nonzero number ϵ and

$$k_3 = \epsilon \left(-\frac{b_3 w}{|w|^2}, 1 \right).$$

Then $k_3 \perp n_3$, its axial component is nonzero, and $\bar{v}_3 \neq 0$. Hence k_1, k_2, k_3 are linearly independent. Writing $\bar{v}_3 = (X, Y)$, take

$$k_4 = (-Y, X, 0), \quad k_5 = 0.$$

The fourth increment is perpendicular to n_4 , and

$$\det(\bar{n}_4, \bar{n}_5) = X^2 + Y^2 > 0.$$

Thus n_4, n_5 are nonparallel. All the normals in this construction are nonzero, and every chosen increment can be represented by $L_i = -n_i \times k_i / |n_i|^2$. Choose $L_5 = 0$.

At the equal-time configuration,

$$x_5 = x_0 - v_0(t_0 - t_1) - \sum_{i=1}^4 v_i(t_i - t_{i+1}).$$

All derivatives of x_5 in angular directions vanish. Keeping t_5 fixed, its first three time columns are

$$\partial_{t_1} x_5 = -k_1, \quad \partial_{t_2} x_5 = -k_2, \quad \partial_{t_3} x_5 = -k_3.$$

Derivatives of the v_i , including derivatives of the position-dependent p_i , are multiplied by zero free-flight durations in this formula. In the final velocity block,

$$\delta_{L_4} v_5 = n_4 \times \delta L_4, \quad \delta_{L_5} v_5 = n_5 \times \delta L_5,$$

because $L_5 = 0$ and the p_i have zero angular position derivatives at equal times. The sum of the planes $n_4^\perp + n_5^\perp$ is $\mathbb{R}^3$. Choose three angular columns spanning this sum. Together with the three time columns above, they give an invertible 6×6 block for (x_5, v_5) : its upper-right block is zero and both diagonal blocks are invertible. Appending the t_5 column and target coordinate gives a nonzero 7×7 minor.

This minor is a nontrivial analytic function on the connected domain. Its zero set has Lebesgue measure zero. For completeness, at any zero of a nontrivial analytic function some derivative of finite order is nonzero; otherwise its Taylor series vanishes on a neighborhood, and analytic continuation through overlapping balls in the connected domain makes the function identically zero. Choosing a derivative of minimal nonzero order puts the point on a regular zero set of a derivative of one lower order. There are countably many such derivatives, and each regular zero set is covered by countably many smooth hypersurface charts. Their union is null. Rank failure is contained in the zero set of our particular minor, so it is null as well. Continuity also shows directly that the witness persists at nearby strictly ordered times. It need not obey any spatial mask.

For the prelimit maps replace p_i by (8.3) and include $w_i \in \mathbb{R}^3$ among the parameters. On compact sets the scales have positive lower bounds, $d_{r_n \mu_i} \rightarrow 0$, and all implicit derivatives have the same positive lower bound one. Differentiating the implicit equations, or applying their implicit function formulas, gives C^1 convergence in the times, positions, and all bounded L_i, w_i . The limit is Φ , independent of the fifteen spectator w_i -coordinates. Its rank is still seven almost everywhere on the enlarged parameter domain, by Fubini. Lemma 8.7 now proves the final assertion. $\square$

Lemma 8.9 (Vanishing of five nonaxial gains). *Suppose $\ell_n \rightarrow 0$, the hit endpoints converge after periodic translation, and their limiting velocity is nonaxial. Suppose also that $0 \leq f_n \leq 2$ on $a \leq t \leq \tau_n$ and $f_n \rightarrow 0$ in measure on every compact subset of*

$$(0, U) \times \mathbb{R}_x^3 \times \{v \in \mathbb{R}^3 : \bar{v} \neq 0\}.$$

Then

$$(\mathcal{V}_n^5 f_n)(\tau_n, x_n, v_n) \longrightarrow 0.$$

Proof. Fix an arbitrarily small error tolerance. Remove the tails of Lemma 8.3. All path velocities and positions are now in fixed bounded sets. At every encountered main core, $s \leq Cr_n$, so its transverse width tends uniformly to zero.

Starting from a positive lower bound for the endpoint transverse speed, apply the nonaxial small-input estimate in Lemma 3.5 successively at gains $1, \dots, 5$. Choose positive numbers $b_1, \dots, b_5$ so that deleting $|\bar{v}_i| < b_i$ costs less than the prescribed error at step i . At each step the previous lower bound has already been fixed; the later gain masses are bounded by (8.6). Thus this is a finite successive choice, followed by $n \rightarrow \infty$. The sum of the five deleted contributions is arbitrarily small. Write $b > 0$ for a lower bound on all retained transverse speeds, including v_0 . For large n , every retained normal $v_{i-1} - p_i$ has length at least $b/2$.

Next delete collision points with $|y_i| < \rho$. Conditional on all earlier parameters, y_i is an affine function of t_i with derivative $\bar{v}_{i-1}$. Consequently

$$|\{t_i : |y_i| < \rho\}| \leq \frac{2\rho}{b}.$$

Integrate the current kernel using its mass bound C/a , and bound the later kernels by (8.6). Summing over the five steps bounds this deletion by $C_a \rho/b$. Choose $\rho > 0$ after the speed

thresholds. The retained scales now belong to a compact subinterval $[\mu_-, \mu_+] \subset (0, \infty)$. Mesh convergence gives a fixed finite list of scales on this interval, with positive limits, after an immaterial enlargement of its endpoints. There are consequently only finitely many five-tuples of scales and signs to consider.

The tested time t_5 must remain inside the weak-limit interval. If $U > 1$, it already lies in the compact interval $[a, 1] \Subset (0, U)$. If $U = 1$, delete $t_5 > 1 - \delta$. Ordering forces all five times into $(1 - \delta, \tau_n)$, whose length is at most δ . This costs at most

$$2(C/a)^5 \frac{\delta^5}{5!}. \quad (8.16)$$

Choose $\delta > 0$ to make this as small as desired. No convergence at the event time is being asserted.

Fix one retained scale/sign tuple. Drop all spatial masks in constructing its maps, and retain the masks in its nonnegative integrand. Each plane increment is bounded, all normals are bounded above and below, and each cold parameter is bounded. Apply Lemma 8.6 successively. The integral becomes an integral over five times, fifteen L -coordinates, and fifteen cold parameter coordinates. Its support lies in a fixed compact subset of $(0, \infty)^5 \times \mathbb{R}^{30}$, and its density is uniformly bounded: the core amplitudes are at most C/a , the factors $|n_i| \zeta(L_i \cdot n_i / |n_i|)$ are bounded, and all retained ratios $H(v_i)/H(v_{i-1})$ are bounded. The masks, time ordering, and cut conditions are arbitrary measurable factors between zero and one in this density.

The target (t_5, x_5, v_5) on the actual support lies in a fixed compact set B inside the open weak-limit region, away from the velocity axis. Set $F_n = f_n \mathbf{1}_B \mathbf{1}_{\{a \leq t \leq \tau_n\}}$ and extend it by zero; it is bounded by two and tends to zero in measure. The maps converge in C^1 to the map in Lemma 8.8. That lemma and Lemma 8.7 show that the integral for this tuple tends to zero. Notice that the entire parameter compact need not map into B ; only the support of the measurable density must do so.

There are finitely many tuples, so their sum tends to zero. The total limsup of the original integral is bounded by the arbitrarily small errors from the preceding deletions. Letting those tolerances tend to zero proves the lemma. $\square$

8.6 Completion of the nonvanishing argument

Completion of the proof of Proposition 8.1. Assume $l = 0$ and $g = 0$, as in the reduction above. Choose endpoints (x_n, v_n) at time τ_n in the common full-measure set fixed above such that $f_n(\tau_n, x_n, v_n) \geq 1/2$. For each fixed a , the residual in (8.8) tends to zero along any subsequence of these endpoints. Indeed, Lemma 8.3 first excludes large velocities and escaping positions. Otherwise choose convergent endpoints after a periodic translation. If the limiting velocity is axial, the first two gains vanish by Lemma 8.5, and the last three have bounded mass by (8.6). If it is nonaxial, use (8.7) in Lemma 8.9. This covers every subsequence; hence the residual tends to zero for the original endpoint sequence as well.

Taking the limsup in (8.8) at these endpoints gives

$$\frac{1}{2} \leq a \sum_{j=0}^4 \frac{(C \log(1/a))^j}{j!}.$$

Finally let $a \downarrow 0$. The right side tends to zero, a contradiction. Throughout this argument U was allowed to be infinite, and the case $U = 1$ was handled by (8.16). This proves the proposition. $\square$

9 Branching growth and the dilated horizon

We now choose the parameters left free in the construction. Their choice will be independent of the subsequence in Proposition 7.1, including the phase of its geometric mesh and its lower endpoint $l \in [0, 1]$.

Proposition 9.1. *There are fixed positive choices of η, α, γ, K , with η, α, γ small and K large, for which every limit in Propositions 7.1 and 8.1 has $U < \infty$. These parameters can be fixed before the final choices of s_0, d_0, T_0 . The same choices work for every limiting mesh and every $l \in [0, 1]$.*

The proof uses only positive, finite collision histories starting at a strictly positive time. We first establish estimates in a constant medium.

9.1 Positive histories and their normalization

For $v, w \in \mathbb{R}^3$ and $n \in S^2$, set

$$\Phi(v, w, n) = \frac{v + w}{2} + \frac{|v - w|}{2}n, \quad d\varsigma(n) = \frac{dn}{4\pi}. \quad (9.1)$$

Thus ς is a probability measure. By Lemma 3.1, a collision with incident velocities v, w has rate $2\pi|v - w|$, and the marginal law of either child is the pushforward of ς under $\Phi(v, w, \cdot)$. The two marginal laws agree because $n \mapsto -n$ preserves ς . Consequently, for a nonnegative velocity measure B and a density f ,

$$\int_{\mathbb{R}^3} P(B, f)(v) dv = \int_{\mathbb{R}^3} \nu_B(v) f(v) dv. \quad (9.2)$$

The coefficient 2 in the gain of (7.5) therefore counts both children, while the loss removes their parent once.

More generally, fix Borel representatives of the coefficients of a two-beam medium

$$\tilde{B}(t, x) = \sum_{\sigma=\pm 1} b_\sigma(t, x) \delta_{w_\sigma(t, x)}, \quad q_\sigma(t, x, v) = 2\pi b_\sigma(t, x) |v - w_\sigma(t, x)|, \quad \nu = \sum_{\sigma} q_\sigma.$$

All pointwise path conditions below refer to these representatives. On $[a, b]$, a history with ordered collision times $a < \tau_1 < \dots < \tau_j < b$, signs σ_i , and directions n_i follows free flights between collisions and uses

$$V_{\tau_i} = \Phi(V_{\tau_i-}, w_{\sigma_i}(\tau_i, X_{\tau_i}), n_i).$$

Its weight relative to $\prod_i d\tau_i d\varsigma(n_i)$ is

$$\exp\left(-\int_a^b \nu(s, X_s, V_s) ds\right) \prod_{i=1}^j 2q_{\sigma_i}(\tau_i, X_{\tau_i}, V_{\tau_i-}). \quad (9.3)$$

For $j = 0$ the empty product is one.

Lemma 9.2. *Suppose $f \geq 0$ solves $D_t f = 2P(\tilde{B}, f) - \nu f$ in the local transport sense. On bounded positive-time regions where the coefficients and collision terms are integrable, start with any nonnegative portion of $f(a)$. Integration of any collection of bounded finite histories with weights (9.3) gives a measure dominated by $f(b)$.*

Proof. Set $G = 2P(\tilde{B}, f)$. In free coordinates $f^\sharp(t, x, v) = f(t, x + tv, v)$, the equation becomes

$$\partial_t f^\sharp = (G - \nu f)^\sharp.$$

On bounded tubes its right side is locally integrable, so $f^\sharp$ has absolutely continuous time sections for almost every (x, v) and local L^1 time traces. Fubini's theorem also makes $\nu^\sharp$ integrable on almost every bounded characteristic segment. The one-dimensional integrating-factor formula therefore consists of the freely transported initial density multiplied by its survival factor, plus the time integral of the nonnegative gain with the same survival factor. Repeated substitution into the gain gives (9.3), by (9.1) and (9.2). Every remainder is nonnegative. Restricting the histories, or the initial density, preserves the lower bound. Tonelli's theorem justifies the substitutions. On bounded portions with total gain rate at most C , the j -collision term is bounded by $C^j(b-a)^j/j!$ times the initial mass.

These formulas apply to point-velocity partners. Indeed Lemma 3.1 gives a density for the gain against each such partner, with mass (9.2). Thus free flight and each gain preserve absolute continuity when the starting measure has a density. Null spatial sets in representatives of the coefficients do not affect the integrated formula. $\square$

For the background (7.3), $|u_\sigma| \leq 1 + \alpha$ and

$$0 \leq b_\sigma(t, x) \leq \frac{\theta K}{t\alpha\gamma}, \quad \nu_B(t, x, v) \leq \frac{C(1+|v|)}{t} \quad (t > 0). \quad (9.4)$$

At each mesh interface choose the value from the outer adjacent annulus, and on the transverse axis set the medium to zero. These are Borel representatives obeying the same bounds. All bounded finite flights starting at positive time have strictly positive survival. The cap and Lemma 3.4 give locally integrable collision terms. The local transport formula in Proposition 7.1 therefore supplies time traces of g on bounded regions. Testing nonnegative compact functions and taking a time limit shows that $g(t) \leq (t+l)H$ also holds at these traces. We use this representative.

9.2 A reference flight with a positive potential

We convert the branching history weights into expectations along a single Markov flight with a positive potential. This is a jump-process Feynman–Kac representation, related to the many-to-one formulas for branching systems; see [9, Section 4.1] for general probabilistic context. The finite-history calculation below proves the precise representation for the present collision law.

Fix

$$D = 5, \quad E = \{v : |v| \leq D\}, \quad B_0 = \delta_{e_3} + \delta_{-e_3}, \quad \nu_0(v) = 2\pi(|v - e_3| + |v + e_3|).$$

Let $J(v, dv')$ be the single-child law obtained by choosing a sign with probabilities proportional to $|v - \sigma e_3|$ and then using (9.1). Put $p(v) = J(v, E)$.

Lemma 9.3. *For every $v \in E$, $p(v) \geq 25/26$. The reference history sum restricted to velocities in E is represented by a Markov flight with velocity jump rate $\lambda(v) = 2p(v)\nu_0(v)$, jump law $J(v, dv')\mathbf{1}_E(v')/p(v)$, and multiplicative weight*

$$\exp\left(\int_0^A \mathcal{W}(V_s) ds\right), \quad \mathcal{W}(v) = (2p(v) - 1)\nu_0(v).$$

One may take

$$0 < c_0 = \frac{48\pi}{13} \leq \mathcal{W}(v), \quad \lambda(v) \leq C_0 = 48\pi \quad (v \in E). \quad (9.5)$$

Proof. Set $s = |v|^2$, $z = v_3$, and $R = \sqrt{(s+1)^2 - 4z^2}$. For either partner sign the squared child speed has the law

$$|v'|^2 = \frac{s+1+RZ}{2}, \quad Z \text{ uniform on } [-1, 1].$$

This follows by expanding the square in (9.1): a coordinate of a uniform point on S^2 is uniform on $[-1, 1]$. The maximum is at most $s+1$, so there is no exit for $s \leq 24$. For $24 < s \leq 25$, the exit probability, if positive, is

$$\frac{1}{2} + \frac{s-49}{2R} \leq \frac{1}{2} + \frac{s-49}{2(s+1)} = \frac{s-24}{s+1} \leq \frac{1}{26}.$$

The inequality uses $R \leq s+1$ and $s-49 < 0$. If $R = 0$, the squared speed is deterministic and at most 25. Mixing the signs preserves the bound.

A Markov history of rate λ has the survival factor $e^{-\int \lambda}$, and one rate times jump-law factor per collision. Multiplication by $e^{\int (\lambda - \nu_0)}$ changes this exactly to (9.3), restricted to E . Both history sums converge absolutely on finite intervals because their rates are bounded. Finally $4\pi \leq \nu_0(v) \leq 24\pi$ on E , and $2p(v) - 1 \geq 12/13$, giving (9.5). $\square$

9.3 Three collisions give a volume minorization

Write $P_t(v, \cdot)$ for the transition law of the preceding velocity process. We establish a Doeblin minorization, the classical common-mass condition for mixing of Markov chains [5]. The three-collision geometry proving this condition and the resulting contraction argument are both given below.

Lemma 9.4. *There are $r_0 > 0$ and $\varepsilon_0 > 0$ such that*

$$P_1(v, A) \geq \varepsilon_0 |A \cap B(0, r_0)| \quad (v \in E) \quad (9.6)$$

for every Borel set $A \subset E$. For each $\zeta > 0$, there is also $\varepsilon_\zeta > 0$ such that

$$P_1(v, B(e_1, \zeta) \cap \{|v'| < D\}) \geq \varepsilon_\zeta \quad (v \in E).$$

Proof. Given $v \in E$, choose a sign with $|v - \sigma e_3| \geq 1$. Use this partner for the first collision, and select an output within a small fixed distance δ of σe_3 . Its probability has a uniform positive lower bound: the incident distance lies in $[1, 6]$, and a cap of fixed Euclidean radius δ at that endpoint has area probability bounded below uniformly on the corresponding diameter spheres. Such first outputs lie strictly in E .

Use partner $-\sigma e_3$ next. If the first output is exactly σe_3 , the second sphere is the unit sphere; restrict its output u near e_1 . For the third collision use partner e_3 . The last output is

$$T(u, n) = \frac{u + e_3}{2} + \frac{|u - e_3|}{2} n.$$

At $u = e_1$, $n_0 = -(e_1 + e_3)/\sqrt{2}$, this output is zero. The two angular derivatives in n span $(e_1 + e_3)^\perp$. The vector e_3 is tangent to the preceding unit sphere at e_1 , and direct differentiation gives

$$\partial_{e_3} T(e_1, n_0) = \frac{e_1 + 3e_3}{4}, \quad (e_1 + e_3) \cdot \partial_{e_3} T(e_1, n_0) = 1. \quad (9.7)$$

Thus the map of the four angular coordinates in the last two collisions to the last velocity has rank three.

Take the two coordinates of n and the coordinate of u with derivative e_3 as distinguished variables, and hold the fourth coordinate in a short interval. The inverse function theorem, with this fourth coordinate and the first output as parameters, gives neighborhoods on which the distinguished-variable map is one-to-one, its Jacobian and inverse Jacobian are bounded, and its image contains a fixed ball $B(0, r_0)$. Shrink these neighborhoods and δ so this holds for every first output within δ of σe_3 , for both signs. Sphere area in these coordinates has a positive lower density. Change of variables in the three distinguished coordinates, and integration in the fourth, give a fixed positive multiple of volume on $B(0, r_0)$.

Force the three times into three fixed, disjoint intervals inside $(0, 1)$, and prohibit other jumps. The first selected relative speed is at least 1, the second is close to 2, and the third close to $\sqrt{2}$. In the reference Markov process the selected transition intensity is

$$4\pi|v - \sigma e_3| \mathbf{1}_E(\Phi(v, \sigma e_3, n)) \, d\zeta(n).$$

It has positive uniform lower density on the patches just specified. The no-additional-jump factor is at least e^{-C_0} . Integrating the times proves (9.6). Keeping only the first two collisions, selecting a small patch near e_1 in the second, and prohibiting later jumps proves the last assertion. $\square$

Lemma 9.5. *The reference process has a stationary probability law π with $\int v \, d\pi(v) = 0$. There are $C, c > 0$ such that*

$$\sup_{v \in E} \|P_t(v, \cdot) - \pi\|_{\text{TV}} \leq C e^{-ct}, \quad \sup_{v \in E} \mathbb{E}_v \left| A^{-1} \int_0^A V_s \, ds \right|^2 \leq \frac{C}{A} \quad (A \geq 1).$$

Proof. The normalized measure on the right side of (9.6) is a common part of positive mass ϵ in all rows of P_1 . Removing this common part shows that the map $\mu \mapsto \mu P_1$ contracts total variation distance by $1 - \epsilon$. Probability measures form a complete space in that metric, so iteration yields a unique fixed point π and uniform geometric convergence. Since P_t commutes with P_1 , πP_t is another fixed point of P_1 , hence equals π . Contractivity of Markov kernels gives the estimate between integer times.

Negation of velocity interchanges the two signs and preserves E , so uniqueness of π makes it symmetric and its mean zero. Consequently $\sup_{v \in E} |\mathbb{E}_v V_t| \leq C e^{-ct}$. The Markov property and bounded velocities imply

$$|\mathbb{E}(V_s \cdot V_t)| \leq C e^{-c(t-s)} \quad (0 \leq s \leq t)$$

uniformly over starting states. Integrating this estimate on $[0, A]^2$ bounds the second moment of $\int_0^A V_s \, ds$ by CA . $\square$

9.4 A displacement block and measurable perturbations

Fix $d = 1/10$, and choose once and for all

$$0 < \kappa < \min \left\{ \frac{1}{4}, \frac{c_0}{2C_0}, \frac{r_0}{4} \right\}. \quad (9.8)$$

In particular a small ball about κe_1 lies in the minorization ball.

To repeat growth in the cold background, we will transport mass through boxes whose transverse distance from the axis grows proportionally to time:

$$\mathcal{B}_t = \{(x, v) : |x - \kappa t e_1| \leq d\kappa t, |v| \leq D\}, \quad t > 0.$$

For a duration $\Delta > 0$, a displacement within $d\kappa\Delta$ of $\kappa\Delta e_1$ carries every starting point of $\mathcal{B}_t$ into $\mathcal{B}_{t+\Delta}$, provided the final velocity remains in E . We first obtain growth with this displacement in the reference medium, then transfer it to the varying cold background.

Lemma 9.6. *There is a finite L such that, for all*

$$A \in I_L := \left[\frac{L}{1+d}, \frac{L}{1-d} \right], \quad v \in E,$$

reference branching histories with velocities in E , final velocity strictly inside E , and displacement Z_A satisfying

$$|Z_A - \kappa A e_1| < \frac{3}{4} d\kappa A \tag{9.9}$$

have total weight greater than 8.

Proof. Put $A_0 = (1 - \kappa)A - 1$. On the first interval of length A_0 , require the reference Markov flight to have integrated velocity of norm at most $Ad\kappa/4$. Lemma 9.5 and Chebyshev's inequality give probability at least $1/2$, uniformly in the start when A is sufficiently large. Over the next unit interval require the last velocity to lie in $B(e_1, d/4)$. The conditional probability is at least a fixed $\varepsilon > 0$, by Lemma 9.4. Require no jump during the final interval of length κA , whose conditional probability is at least $e^{-C_0\kappa A}$.

The errors in displacement from $\kappa A e_1$ on the three intervals are at most $Ad\kappa/4$, D , and $Ad\kappa/4$. For $A > 4D/(d\kappa)$ their sum is less than $3Ad\kappa/4$. The final velocity lies strictly in E . Lemma 9.3 bounds the branching weight below by

$$\frac{\varepsilon}{2} \exp((c_0 - C_0\kappa)A).$$

Its exponent is positive by (9.8). Choose L large enough that this exceeds 8, and all the preceding lower bounds on A hold throughout I_L . $\square$

Lemma 9.7. *For these fixed κ, L , there are $J \in \mathbb{N}$ and $\varepsilon_* \in (0, 1/4)$ with the following property. Suppose a measurable medium obeys, throughout every speed- D excursion from a specified starting point over a duration $A \in I_L$,*

$$|b_\sigma - 1| \leq \varepsilon_*, \quad |w_\sigma - \sigma e_3| \leq \varepsilon_* \quad (\sigma = \pm 1). \tag{9.10}$$

From each starting velocity in E , histories with at most J collisions, speed at most D on every free flight, every postcollision velocity and the final velocity strictly inside E , and

$$|Z_A - \kappa A e_1| < d\kappa A \tag{9.11}$$

have total weight greater than 3. The initial free flight may have speed D . No continuity of the medium is needed.

Proof. Fix a reference start and duration $(v, A) \in E \times I_L$. The zero-collision term has weight $e^{-\nu_0(v)A} \leq 1$. Discarding it from the family in Lemma 9.6 leaves weight greater than 7. Monotone convergence in the positive history count permits restricting to finitely many counts $j \geq 1$ while retaining weight greater than 7. Every postcollision velocity lies strictly in E almost surely: the intersection of a nondegenerate diameter sphere with ∂E has sphere area zero, and a degenerate collision has zero rate. This includes the final velocity. No inward margin is required for the initial velocity. The displacement inequality is strict. Ordered times may be kept apart and away from the endpoints. Using finitely many sphere coordinate charts and inner regularity of their parameter integrals, restrict further to compact parameter sets with total weight greater than 6 and positive margins in all these restrictions, including $|Z_A - \kappa A e_1| < d\kappa A$. Parameterize the times as fractions of A .

The reference paths and weights are continuous in these finite parameters and in (v, A) . The fixed compact tests therefore retain weight greater than 5 and at least half their original positive margins in a sufficiently small relative neighborhood of that start and duration in $E \times I_L$. These neighborhoods cover the compact set $E \times I_L$. A finite subcover gives finitely many tests, a maximum count J , and uniform positive margins.

Use the same initial velocity, signs, directions, and fractional times for each perturbed history and its reference history. Their first flights have speed at most D , including boundary starts. The sphere map satisfies

$$|\Phi(v, w, n) - \Phi(\tilde{v}, \tilde{w}, n)| \leq |v - \tilde{v}| + |w - \tilde{w}|.$$

Induction bounds the velocity error after j collisions by $j\varepsilon_*$ and the displacement error by $AJ\varepsilon_*$. This comparison does not compare the medium at two nearby positions: its value at the actual perturbed position is close to the same constant reference partner. A sufficiently small ε_* preserves all the strict velocity and displacement margins. The actual paths thus have speed at most D , so they stay within the region on which closeness was assumed. Formally this follows by induction up to the first possible failure; the preserved margins exclude such a failure.

The relative-speed errors and full loss-rate errors on these histories are bounded by $C_J\varepsilon_*$. The density error gives the same bound for the gain-rate factors. The interval has bounded length, so the error in the integrated loss and its exponential is bounded by $C_{J,L}\varepsilon_*$. Finite products of the bounded gain factors then show that (9.3) changes by at most $C_{J,L}\varepsilon_*$ uniformly on each test. The tests have finite parameter measure. Decreasing ε_* makes their integrals greater than 3. All coefficient estimates were pointwise; measurability is sufficient. $\square$

9.5 Uniform choices of the construction parameters

We first reserve one more small geometric tolerance. There is $\alpha_{\text{acc}} > 0$, depending only on κ and Lemma 9.4, with this property: after a first collision puts a velocity sufficiently close to σe_3 , two more collisions, with partners within α_{acc} of $-\sigma e_3, e_3$, have positive angular probability of ending in

$$B(\kappa e_1, d\kappa/4). \tag{9.12}$$

Indeed the inverse-coordinate construction in that lemma, with first output exactly σe_3 , maps a positive-measure angular set into $B(\kappa e_1, d\kappa/8)$, which is inside $B(0, r_0)$. Keep a compact subset of positive measure in this set. Its outputs have a positive distance from the complement of (9.12); the intermediate velocities and relative speeds remain in the neighborhoods of that construction. Uniform continuity of the two sphere maps preserves this for sufficiently small

perturbations of the first output and partners, including measurable partners at the actual path positions. Take the smaller tolerance for the two possible signs. We also require α_{acc} to be less than half the permitted radius of the first-output neighborhood.

The constants $\kappa, L, J, \varepsilon_*, \alpha_{\text{acc}}$ are now fixed and do not involve any construction parameter. Fix preliminary bounds $\eta \leq 1, \alpha \leq 1, K \geq 1$. For $t > 0$, put

$$\Delta = \frac{L\kappa t}{\theta K}, \quad a = \frac{L\kappa}{\theta K}. \quad (9.13)$$

At a starting point $x_0 = (y_0, z_0)$ in $\mathcal{B}_t$,

$$(1-d)\kappa t \leq |y_0| \leq (1+d)\kappa t.$$

Every speed- D excursion over $[t, t + \Delta]$ satisfies

$$||y| - |y_0|| \leq D\Delta, \quad |z| \leq d\kappa t + D\Delta.$$

Set $\delta_K = DL/(\theta K(1-d))$. By increasing K , impose $\delta_K \leq 1/4$ and $a \leq 1/4$. Then

$$|y| \geq (1 - \delta_K)(1 - d)\kappa t.$$

For all sufficiently large t , every such excursion is outside the lower mesh hole, since $l \leq 1$. On its active annulus $e^{-\eta}|y| \leq \mu \leq |y|$, for every mesh phase. Thus, with a fixed constant C_1 depending only on d, D ,

$$\begin{aligned} \left| \frac{|y_0|}{\mu} - 1 \right| &\leq C_1(\eta + \delta_K), \\ \frac{\tau}{\mu} &\leq \frac{C_1}{\kappa}, \\ \frac{|z - \tau u_\sigma|}{\mu} &\leq \frac{C_1}{\kappa} \quad (t \leq \tau \leq t + \Delta). \end{aligned} \quad (9.14)$$

The last estimate uses $|u_\sigma| \leq 2$.

Use the reference density factor $\rho_0 = \theta K/|y_0|$, and multiply both time and position units by ρ_0 ; velocities are unchanged. The reference duration is exactly

$$A = \rho_0 \Delta = \frac{L\kappa t}{|y_0|} \in I_L. \quad (9.15)$$

With $S_\sigma = S(\gamma(z - \tau u_\sigma)/\mu)$, the density ratio in these units is

$$\frac{b_\sigma(\tau, x)}{\rho_0} = \frac{|y_0|}{\mu} \frac{S_\sigma}{1 + (\tau\alpha\gamma/\mu)S_\sigma}. \quad (9.16)$$

Because $1 - \text{sech}^2 h = \tanh^2 h \leq h^2$, (9.14) gives, when the right-hand small quantities are at most one,

$$\left| \frac{b_\sigma}{\rho_0} - 1 \right| \leq C_2 \left(\eta + \delta_K + (\gamma/\kappa)^2 + \alpha\gamma/\kappa \right), \quad |u_\sigma - \sigma| \leq \alpha. \quad (9.17)$$

Here C_2 is independent of t, l , and mesh phase.

Choose $\eta > 0$ small, then

$$0 < \alpha < \min\{1/4, \varepsilon_*, \alpha_{\text{acc}}\},$$

then $\gamma > 0$ small, so that the first, third, and fourth terms on the right of (9.17), including their factor C_2 , sum to less than $\varepsilon_*/2$. Finally increase K so the δ_K term is less than $\varepsilon_*/2$, the preliminary restrictions hold, and

$$(1 + a)^4 < 3. \quad (9.18)$$

This establishes (9.10) for all the scaled blocks, with no choice depending on a limit, mesh, or solution.

These choices precede the final small geometric and thermal parameters. Making $\alpha\gamma$ small may increase the fixed coefficient in (9.4) and earlier comparison bounds, but introduces no dependence on the approximation index. Propositions 4.4 and 5.1 allow s_0, d_0 , and then the final short horizon T_0 , to be chosen after these fixed constants. We use those choices henceforth.

9.6 Access to a late box and the growth contradiction

Lemma 9.8. *For the fixed parameters above, let $g \not\equiv 0$ solve (7.5) on $0 < t < \infty$ with $0 \leq g \leq (t + l)H$, for any allowed mesh and $l \in [0, 1]$. Then $\int_{\mathcal{B}_t} g(t, x, v) dx dv > 0$ at arbitrarily large finite times.*

Proof. At some positive time a_0 , a bounded portion of $g(a_0)$ has positive mass. The velocity axis has Lebesgue measure zero, so a further portion of positive mass satisfies

$$|x| \leq R, \quad |v| \leq R, \quad |\bar{v}| \geq \delta > 0.$$

For a sufficiently long but finite free flight of length s ,

$$|y + s\bar{v}| \geq s\delta - R > l + 2$$

on this portion. Its survival is bounded below by a positive constant by (9.4). Lemma 9.2 gives positive mass in a bounded region strictly outside the hole at time $a_1 = a_0 + s$.

Choose $h > 0$ small so that paths whose initial speed is at most R and whose later speeds are at most 3 remain in a bounded region with $|y| > l + 1$ during $[a_1, a_1 + h]$. Both sign densities have strictly positive lower bounds there. To see this directly, the time, axial coordinate, and transverse radius are bounded, the transverse radius has a positive lower bound, and $e^{-\eta}|y| \leq \mu \leq |y|$. The argument of S is therefore bounded, its numerator is positive, and the denominator in (7.3) is bounded above. The rates on the selected bounded-speed paths also have finite upper bounds.

Confine three collisions to this short interval. Choose a reference sign with $|v - \sigma e_3| \geq 1$ for the first collision. The actual relative speed is at least $1 - \alpha \geq 3/4$. Choose its output in a sufficiently small cap about the actual partner. The cap probability has a positive lower bound for inputs bounded by R . Since the actual partner is within α of σe_3 , this first output lies in the reserved first-output neighborhood. Next use the signs $-\sigma$ and $+1$, and the compact angular tests defining α_{acc} . Their final velocity lies in (9.12). The second output stays near e_1 , so the second and third relative speeds have positive lower bounds; all postcollision speeds are at most 3. Put the three times in disjoint subintervals and prohibit other collisions. Positive lower bounds for the selected gain factors, angular probabilities, time-interval lengths, and survival give strictly positive total weight. The sphere-map comparison used here requires only uniform partner closeness, and therefore remains valid for measurable coefficients.

There is consequently a bounded positive portion at time $a_2 = a_1 + h$ with $|v - \kappa e_1| < d\kappa/4$. Let R_2 bound $|x - \kappa a_2 e_1|$ on that portion. After free flight to any sufficiently large finite time t ,

$$|x + (t - a_2)v - \kappa t e_1| \leq R_2 + \frac{d\kappa}{4}(t - a_2) < d\kappa t.$$

Its speed is strictly less than D . For each finite t , survival remains strictly positive by (9.4). The zero-collision term of Lemma 9.2 proves the assertion. $\square$

Proof of Proposition 9.1. Suppose an extracted limit has $U = \infty$. Proposition 8.1 makes it nonzero. Lemma 9.8 gives a sufficiently large t_0 with positive mass m_0 in $\mathcal{B}_{t_0}$. Increase t_0 if necessary so that all the excursions considered in (9.13) lie above the lower hole. The same holds for every later block.

At each starting point of $\mathcal{B}_t$, apply Lemma 9.7 using the simultaneous space–time scaling (9.15). Translating back its displacement restriction gives

$$|x_{\text{end}} - x_{\text{start}} - \kappa \Delta e_1| < d\kappa \Delta.$$

The starting radius $d\kappa t$, followed by the triangle inequality, places the endpoint in $\mathcal{B}_{t+\Delta}$. The velocity belongs to E : the selected positive-count histories have a strictly interior final velocity. Their first flight need only have speed at most D , exactly as in the medium comparison. History weights are unchanged by this scaling: each gain rate is divided by ρ_0 , each time differential is multiplied by ρ_0 , and the integrated loss is invariant. The selected histories therefore have total weight greater than 3 uniformly over starts in $\mathcal{B}_t$.

Use the kernel obtained by integrating all histories with $1 \leq j \leq J$ that satisfy the block's velocity and displacement restrictions. The finite compact tests above certify its weight greater than 3; no choice of a test for each starting point is needed. Integrate this positive kernel against the portion of $g(t)$ in the box and use Lemma 9.2. At $t_n = (1 + a)^n t_0$, induction gives

$$\int_{\mathcal{B}_{t_n}} g(t_n, x, v) dx dv \geq 3^n m_0.$$

The cap and $l \leq 1$, on the other hand, give

$$\int_{\mathcal{B}_{t_n}} g(t_n, x, v) dx dv \leq (t_n + 1) \frac{4\pi}{3} (d\kappa t_n)^3 \int_E H(v) dv \leq C(1 + t_n)^4.$$

This contradicts (9.18) as $n \rightarrow \infty$. The positive mass m_0 and starting time may depend on the particular limit; their sizes are immaterial to this comparison. The parameters governing the blocks were fixed before the mesh and l were chosen. Thus every extracted limit has finite dilated horizon. $\square$

10 Separation of two solutions with the same datum

We now combine the two compactness procedures. All parameters have been fixed as in Proposition 9.1, with the final small parameters chosen to satisfy the preceding cold and cap estimates. Fix one dormant limit F^d on $[0, T_0]$, supplied by Proposition 6.1.

Lemma 10.1. *As $T \downarrow 0$,*

$$T^{-1}F^d(Tt, Tx, v) \longrightarrow 0$$

in distributions on compact subsets of

$$0 < t < 1, \quad x \in \mathbb{R}^3, \quad v \in \mathbb{R}^3, \quad \bar{v} \neq 0.$$

The evaluation in x uses the periodic lift centered at the origin.

Proof. The dormant majorant passes to the fixed-horizon weak limit. Its hot part is at most $t_{\text{old}}^2 H$, so after dilation and multiplication by T^{-1} it is bounded by $Tt^2 H$, which vanishes in L^1 on the indicated compact sets.

For the cold majorant, a main enlarged annulus at physical transverse position $y_{\text{old}} = Ty$, with y bounded, has $s \leq CT$. The relevant periodic copy is the one centered at the origin for all sufficiently small T ; other copies contribute only escaped tails. For $|\bar{v}| \geq \varepsilon > 0$, the transverse Gaussian in the cold estimate gives, after enlarging fixed constants,

$$c_{\text{main}}(Tt, Tx, v) \leq Cs^{-31} \exp\left(-\frac{c\varepsilon^2}{d_0^2 s^{20}}\right).$$

The right side, maximized over $0 < s \leq CT$, tends to zero faster than any power of T . At points with no main color only tails are present. The escaped sum is at most $C_A(Tt)^A e^{-4|v|^2}$, and after multiplication by T^{-1} it vanishes for $A > 1$. These uniform bounds on the compact set prove the claim. $\square$

Proof of Theorem 1.1. The datum is (4.4); Lemma 4.2 verifies all its integrability requirements and bounded velocity support. For every horizon $0 < T \leq T_0$, Proposition 5.1 gives cap approximations. Proposition 6.1 gives at least one admissible limit on $[0, T]$, with the same datum as F^d .

Suppose, to obtain a contradiction, that there are horizons $T_k \downarrow 0$ at each of which a cap limit can equal F^d on that interval. At each fixed T_k , choose approximations along a sequence converging to this particular common limit. Use a countable exhaustion by nonnegative smooth compact tests of the open region in Lemma 10.1. A diagonal choice of the approximation indices N_k makes $s_{N_k}/T_k \rightarrow 0$ and gives

$$f_k(t, x, v) := T_k^{-1} F_k(T_k t, T_k x, v) \longrightarrow 0 \tag{10.1}$$

distributionally throughout that region. Here F_k denotes the selected finite approximation. For completeness, a fixed rescaled test at a fixed T_k is simply a smooth physical test with finite, possibly large norm. Weak convergence at that horizon permits its error to be made arbitrarily small. One then enforces finitely many such errors and a sufficiently small s_N/T_k simultaneously. No convergence rate uniform in T_k is being assumed.

Choose a cap event of $F_k = c_k + h_k$ and write $r_k = t_{*,k} + s_{N_k}$. Proposition 7.1 gives, after subselection,

$$\frac{s_{N_k}}{r_k} \rightarrow l, \quad \frac{T_k}{r_k} \rightarrow U \in [1, \infty],$$

and a dilated hot limit g on $0 < t < U$. Proposition 8.1 says that g is nonzero. This statement supplies an observable lower bound. Since g is a nonnegative density and the velocity axis has zero measure, there are $\psi \in C_c^\infty((0, U) \times \mathbb{R}_x^3 \times \{\bar{v} \neq 0\})$, $0 \leq \psi \leq 1$, and $\delta > 0$ such that

$$\int \psi g > 2\delta, \quad \int \psi(t, x, v) r_k^{-1} h_k(r_k t, r_k x, v) dt dx dv \geq \delta$$

for all sufficiently large k . The second inequality follows from the local weak convergence in Proposition 7.1. The test and lower bound may depend on the extracted limit; no uniform numerical value is needed. Proposition 9.1 says that $U < \infty$. Consequently $\rho_k = r_k/T_k \rightarrow \rho = 1/U > 0$.

Since $F_k \geq h_k$,

$$0 \leq r_k^{-1} h_k(r_k t, r_k x, v) \leq \rho_k^{-1} f_k(\rho_k t, \rho_k x, v).$$

For any compact set with $0 < t < U$, bounded x, v , and $\bar{v} \neq 0$, its image under the right-hand dilation lies in one fixed compact subset of the region in (10.1), for all large k . Positivity allows a single smooth nonnegative test, equal to one on this larger compact set, to dominate the changing tests. The right side therefore tends to zero in local L^1 . It follows that $g = 0$ off the velocity axis. Since $0 \leq g \leq (t + l)H$ is a density and the axis has Lebesgue measure zero, $g = 0$ almost everywhere. This contradicts Proposition 8.1.

Thus for all sufficiently small horizons no cap limit can equal the fixed dormant limit. Choose one such horizon and one cap limit. The two early solutions have the same datum and differ on a set of positive space-time-velocity measure. Proposition 6.1 continues both globally, preserving Definition 2.1. Their early difference persists, which proves the theorem. $\square$

The separation occurs at finite velocities near the initial spatial singularity. The conclusion requires neither an incoming flux from infinite velocity nor an alteration of the conservation and entropy conditions.

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